

Leveraging Machine Learning and Wearables Technology for Sepsis Prediction and Workflow Optimization

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Abstract

This paper investigates the viability of using wearable technology to collect continuous time-series vital sign data from patients on Danish hospital general wards, applying supervised machine learning (ML) to predict sepsis onset within a given time horizon. This directly addresses the hospital challenge of balancing resource expenditures of nurse labor and bedside telemetry with patient risk of complications, a trade-off that currently results in a deadly monitoring gap on Danish hospital wards.

Employing a dual-movement analytical strategy, the study first evaluates machine learning estimators against a proxy of the current Danish clinical standard, which is based on periodic manual checks. The results show that the main wearable-proxy estimator is better at predicting sepsis onset than the clinical baseline proxy, achieving a 67% improvement in sepsis-specific discrimination. Looking at the methodological choices made for the main estimator, the results suggest: [1] This paper's stacked ensemble architecture, with etiology-grouped base-learners, contributes unique predictive value in sepsis prediction. [2] The higher sampling frequency promised by wearables significantly improves prediction and timeliness of treatment. [3] The collection of engineered features, both derived from existing studies and independently conceptualized, provides substantial discriminative value over raw vital signs.

The technical results also reveal a challenge, the severe class imbalance of the dataset-cohort, and the somewhat limited availability of signal, results in severe false positive rates (FPR), risking alarm fatigue. To interrogate the viability of integration at Danish hospitals, the second analytical movement introduces a contextualized Unified Theory of Acceptance and Use of Technology (UTAUT) framework to explore the organizational viability of the system. The analysis establishes the continued need for a human-in-the-loop, demonstrating that treating the system as pure automation would trigger alarm fatigue and trust erosion. Instead, designing for Hybrid Intelligence, the paper proposes a system that informs rather than automates clinical decision-making. By deploying a tiered risk-score interface that mimics the logic and interaction of existing clinical systems, minimizing cognitive overload and the behavioral pathology of disuse.

Based on both technical predictive ability and organizational alignment, the paper ultimately finds a strong case for continued research and potential pilot testing regarding the application of wearable technology and machine learning in sepsis prediction.

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Introduction

Sepsis is a life-threatening medical condition where the body's immune system overreacts to an infection, mistakenly attacking its own healthy tissues and potentially causing organ failure and death. It remains one of the most significant contributors to hospital mortality, accounting for roughly one in five deaths globally (Rudd et al., 2020) and approximately 35% of all immediate causes of death in U.S. acute care hospitals (Rhee et al., 2019). While escalation can typically be prevented through the timely administration of antibiotics and hydrating fluids (Larsen, 2023), efficacy depends entirely on early detection. Because every hour of delayed treatment increases sepsis mortality risk by 7.6% (Kumar et al., 2006), rapid identification is the foundation of sepsis care. However, while Intensive Care Units (ICUs) utilize continuous bedside telemetry, general wards largely rely on periodic manual checks—often every four hours—creating a deadly monitoring gap where deteriorating patients are routinely missed.

Recent advances in wearable sensor technology offer a potential tool to close this gap. Commercially available devices (e.g., Apple Watch, Everion, VitalPatch) can now continuously measure vital signs with near-clinical precision (Haveman et al., 2022; Corliss, 2025). Concurrently, a substantial body of literature proves that machine learning (ML) models can predict sepsis with high accuracy (Moor et al., 2021; Barton et al., 2019). Deploying such systems on the general ward promises to significantly increase sampling frequency while lowering the labor burden of manual track-and-trigger rounds (Prgomet et al., 2016).

Yet, predictive accuracy alone is insufficient. A model that flags deterioration must successfully translate that mathematical signal into a clinical response. This depends entirely on the frontline staff who must trust the alerts, the workflows the alerts enter, and the organizational systems that surround both. While ML predictive capabilities are well-documented, literature exploring the organizational translation of wearable-driven sepsis prediction remains virtually non-existent.

Therefore, this paper proceeds in two analytical movements: evaluating both the mathematical limits of predicting sepsis via wearable-grade data, and the socio-technical feasibility of implementing such a system within Danish hospitals.

| Research Question

Can data collected by modern wearable devices be used to predict sepsis onset on hospital wards in Denmark, and which organizational conditions effects implementation viability?

| Research Objective

This paper investigates whether continuous wearable monitoring can effectively predict sepsis on general wards and evaluates the socio-technical conditions required for its implementation. The research proceeds in two analytical movements. First, it trains a machine learning ensemble on wearable-proxy data to test whether non-invasive vital signs can predict sepsis onset at a clinically useful level compared to current standards. Second, it applies a contextualized UTAUT framework to assess the organizational and economic viability of deploying this predictive system in Danish hospitals. The intended contribution is to bridge predictive machine learning performance with practical implementation viability, two areas the existing literature has so far addressed only in isolation. This will allow the paper to serve as a conceptual exploration, able to inform decision makers in the Danish healthcare system about the technical and organizational promise for using wearables in sepsis prediction.

| Delimitation

This paper explicitly delimits its scope across clinical, technical, and organizational boundaries. Clinically, the focus is restricted entirely to predicting sepsis onset, as defined by the sepsis-3 definition, a choice driven by its unique complex-adaptive physiology and economic burden (Rudd et al., 2020) (detailed in the Background and Rationale). Technically, the machine learning models are strictly constrained to six non-invasive vital signs capturable by current wearable sensors and three context features, deliberately excluding laboratory results and physician notes in training, despite the larger feature set available in the MIMIC-IV ICU dataset used. This choice was made to proxy the features wearables could potentially capture (Johnson et al., 2023). Finally, the socio-technical evaluation using the UTAUT framework (Venkatesh et al., 2003) is specifically delimited to the context of the Danish single-payer healthcare system, addressing its unique structural nurse shortages (Birk et al., 2024) and budget envelopes (see Analysis of Organizational & Economic Viability).

| Contributions

To bridge the gap between algorithmic capability and practical healthcare implementation, this paper makes four primary contributions

- **Developing a highly performant, wearable-constrained predictive model.** By engineering novel biological and trajectory features to feed an etiology-specific stacked ensemble, the paper demonstrates that machine learning restricted strictly to wearable-proxy data outperform the papers proxy for current clinical standard, along with all other tested estimators.
 - Results suggest the higher observation frequencies enabled by wearables would significantly improve prediction performance, while also enabling life-saving earlier intervention.

- The choice to dissect the training cohort into etiology-specific training sets grouping patients by documented clinical diagnosis codes, showed to successfully capture etiology-specific signals. This is a unique contribution, which to the best of my knowledge has not been done before.
- The attempt to proxy physiological features not directly capturable by the raw vital-signs resulted in the creation of a small number of engineered features (such as Fever-Driven Tachycardia) novel to this paper, which proved to provide independent predictive value.
- **Redefining the clinical monitoring bottleneck.** By achieving apparent predictive superiority, using fewer raw feature than the clinical standard, the results suggests that the current limit on ward-level sepsis detection is not a lack of physiological data, but the inability of rule-based systems to transform those signals.
- **Bridging algorithmic prediction with socio-technical viability.** Using a contextualized UTAUT framework, the paper moves beyond isolated technical evaluation to assess the behavioral and organizational considerations.
- **Proposing a design heuristic for Hybrid Intelligence.** Translating the socio-technical findings into practice, it is proposed to deploy the algorithm's output through a tiered, informing risk-score interface that mirrors existing nursing cognitive routines. Combined with recommendations for new clinical governance protocols and intellectual skill training, a framework that empowers frontline clinicians to interrogate the machine's reasoning and safely apply their own contextual knowledge is proposed, ensuring the technology functions as a viable clinical asset rather than a source of alarm fatigue.

Background and Rationale

Of the many complications patients may suffer from, this paper focuses on sepsis as the primary use case for wearable-based machine learning prediction. This selection was deliberate, based on two criteria: [1] Significance of the challenge and [2] suitability for machine learning and wearables application.

| Challenge

This paper uses the Danish single-payer healthcare system as reference, because capped budgets convert prevention into a clear saving. In contrast, in US fee-for-service architecture, hospital revenue scales with billable activity, so the preventing complications can erode top-line income, relying on complicated offset mechanisms such as the Hospital Readmissions Reduction Program, which docks up to 3 percent of Medicare base operating payments and Hospital Value-Based Purchasing based on performance (CMS, 2024). The Danish model largely removes that complexity and revenue tension. Hospitals operate within capped regional

budgets in which block-grant financing covered roughly 83 percent of hospital revenue in 2021, with activity-based DRG (Diagnosis Related Groups) payments redistributing money inside the cap rather than expanding it (Birk et al., 2024; Quentin et al., 2023). Further, Denmark faces a substantial shortage of hospital nurses (4.700 in 2022), suggesting high marginal value gain from reclaimed nursing capacity (Birk et al., 2024).

As covered in the introduction, sepsis-related deaths account for roughly 35% of hospital deaths, and successful treatment is dependent on early detection and treatment (Rhee et al., 2019; Kumar et al., 2006). The capacity for early in-hospital detection, however, is fundamentally stratified by patient location. In the ICU, patients are continuously monitored by bedside telemetry that records vitals such as heart rate, oxygen saturation, and blood pressure in near-real-time, with sharp deviation from baseline automatically generating an alarm, and increased nurse-observation frequency (Vincent et al., 2018). This level of surveillance is necessitated by the poor state of ICU patients, and the multitude of complications they risk facing, and naturally carries added resource expenses for the hospital, namely bedside telemetry equipment and nursing labor.

Looking at the general ward, we see a very different reality. The current standard for detecting clinical deterioration on Danish hospital wards is TOKS (Tidlig Opsporing af Kritisk Sygdom), a risk-scoring system, in the National Early Warning Score family (NEWS), build on a rule-based threshold logic. The system is used across all five Danish regions (Brabrand et al., 2010; Seymour et al., 2016). Here a bedside nurse scores seven variables against fixed thresholds, namely respiratory rate, oxygen saturation, supplemental oxygen, systolic blood pressure, heart rate, body temperature, and consciousness, with the cadence set by Sundhedsstyrelsen and regional protocol at typically every four hours when the patient is stable (Petersen et al., 2016).

The extent of what the four-hour interval between nurse-rounds misses is substantial. Saab et al. (2021) placed continuous monitoring devices on general ward patients and compared the physiological events captured by those devices against what nursing staff recorded during routine assessments. Of the hypotensive episodes and oxygen desaturation events detected by continuous monitoring, 79% and 82% respectively did not coincide with a nursing assessment and therefore went clinically undetected. This finding illustrates the broader point, that periodic assessments systematically miss acute deterioration on general wards. Further, research on failure-to-rescue, defined as death following a treatable complication, has consistently demonstrated that mortality variation between hospitals is driven not simply by differences in complication rates but by differences in the capacity to detect and respond to complications before they escalate (Ghaferi et al., 2009; Silber et al., 1992).

This missingness and monitoring gap becomes especially concerning that retrospective analysis of 546,028 MIMIC-IV hospitalizations reveals 92.1% of suspected sepsis originate on the general ward, versus 7.1% in

the ICU. While ICU cases mathematically show higher progression to severe sepsis (83.1% vs. 13.3%), this is a structural artifact because deteriorating ward patients are routinely escalated to intensive care.

Figure 0 - Suspected Sepsis by Onset Location

Onset Location	Suspected Sepsis (n)	Share	Severe / Shock (n)	Progression
Ward-Onset	212,774	92.1%	28,283	13.29%
ICU-Onset	16,449	7.1%	13,668	83.09%
Post-ICU Onset	1,799	0.8%	19	1.06%

Figure 0 shows summary statistics from the MIMIC-IV dataset, where The "Suspected Sepsis" and "Severe / Shock" flags were derived programmatically by querying the database for specific clinical documentation codes, utilizing the extraction methodology from the MIMIC Code Repository (Johnson et al., 2018).

This difference in the quality and frequency of monitoring between the ICU and the general ward is a direct result of the prioritization challenge budget-constrained Danish hospitals face, of balancing resource expediency of nurse labor and bedside telemetry with patient risk of complications. Here ICU patients face a higher combined risk of complications, justifying the higher quality and frequency of monitoring. The implication here, is that monitoring routines are weaker on general wards, due to prioritization of resource expediency, not because it is the level of monitoring which is completely optimal.

Looking at the economic side, sepsis is a high-cost, high-volume hospital condition. Availability of specific numbers for Denmark are limited, but, in the USA the average admission ranges from \$16,324 without organ dysfunction to \$38,298 for septic shock, with hospital-onset cases averaging \$51,022 (Paoli et al., 2018), accounting for 9.9% of total US aggregate hospital costs in 2021. While these costs are largely carried by the patient in the USA, they are covered by the hospital budget in Denmark. Conversely, even a conservative estimate of Danish hospital expenses for every uncaught sepsis case accelerating in severity would be sizeable. By this token, every prevented sepsis acceleration case, directly frees up budget and nursing resources for other patients for the hospital.

| Suitability for ML & Wearable Application

The clinical urgency and financial incentive established above would matter little if sepsis were not amenable to the proposed technological approach. Sepsis is structurally well-suited to supervised machine learning. It manifests as coordinated drift across several vital signs hours before clinical recognition (Henry et al., 2015; Nemati et al., 2018), producing a tractable multivariate time-series signal with sufficient class prevalence among hospitalized patients to support supervised training. Standardized Sepsis-3 labels, defined by an acute change in SOFA score together with suspected infection (Singer et al., 2016), and large open cohorts provide the scale and ground truth that supervised models require. The relevant feature space, heart rate, respiratory rate, SpO2, temperature, and blood pressure, maps directly onto wearable sensor modalities. Sepsis detection

is therefore not just a serious clinical challenge with economic significance, it also just so happens to be very well suited technically.

| The Gap

While existing literature extensively covers machine learning for sepsis prediction, prior models overwhelmingly rely on data-rich ICU inputs, like laboratory values and clinical notes, that cannot be captured by wearable devices (e.g., Calvert et al., 2016; Desautels et al., 2016; Fleuren et al., 2020). Ghiasi et al. (2022) demonstrated promising results using wearable-grade signals alone, but predictive accuracy did not yet match the standard of care, and organizational viability remained un-answered. By determining the predictive performance achievable when strictly constrained to wearable-grade vital signs, and evaluating the nuances of organizational viability, this paper addresses that gap.

Theoretical Framework

To execute the dual-movement analytical strategy required by the research question, the theoretical framework is divided into two distinct pillars. It first establishes the machine learning foundations necessary to accurately model complex biological trajectories and subsequently introduces the Unified Theory of Acceptance and Use of Technology (UTAUT) to evaluate the resulting model's organizational viability within a clinical setting.

| Machine Learning Foundations

This subsection introduces the most central theories driving machine learning decisions in this paper.

Stacked Generalization

Wolpert (1992) introduces stacked generalization as a method for combining the outputs of multiple base learners. Each base learner is trained on partitions of the training set, predictions on held-out partitions are collected, and a second-level learner is trained to map those predictions to the true outputs. The second-level learner is described as one that "deduces the biases of the generalizer(s) with respect to a provided learning set" (Wolpert, 1992, p. 2): its task is not to break ties but to recognize the systematic error patterns of the base learners and correct for them. Wolpert positions stacking as a sophisticated alternative to cross-validation's winner-takes-all model selection, where rather than choosing one best base learner, the meta-learner learns how the base learners' errors relate.

Diversity in Classifier Ensembles

Kuncheva and Whitaker (2003) examine whether the intuition that ensemble accuracy is driven by classifier diversity holds up empirically. They test ten formal diversity measures, four pairwise (the Q statistic, correlation, disagreement, and double-fault) and six non-pairwise (entropy of votes, difficulty index, Kohavi-Wolpert variance, inter-rater agreement, generalized diversity, and coincident failure diversity), against ensemble accuracy on benchmark classification problems. Their finding is cautious. Although diversity correlates with accuracy in some cases, "our results raise some doubts about the usefulness of diversity measures in building classifier ensembles in real-life pattern recognition problems" (2003, p. 1). The paper does not refute the diversity intuition, but it removes the assumption that any form of decorrelation, by any measure, automatically improves ensemble accuracy.

Class Imbalance and Resampling

When the positive class represents a small fraction of the training data, classifiers that minimize overall error default to the majority class and fail at the prediction task that motivates them. He and Garcia (2009) divide remedies into two families. Data-level techniques modify the training distribution before the classifier sees it, while algorithm-level techniques modify the classifier's objective function or split criterion to reweight the minority class without changing the data.

As alternative to popular Synthetic Minority Oversampling Technique (SMOTE)(Chawla et al., 2002), Liu, Wu, and Zhou (2009) propose the idea of ensemble-based under-sampling (EasyEnsemble). Rather than discarding most of the majority class through a single random under-sample like in SMOTE, the method draws multiple disjoint majority subsets, trains a classifier on each, and aggregates their predictions, which preserves the full minority class while distributing the majority-class information across the ensemble rather than throwing most of it away. Both sub-families therefore preserve every minority example; they differ in whether they create new minority points by interpolation or recover the discarded majority signal through aggregation.

Evaluation under Skewed Prevalence

Saito and Rehmsmeier (2015) argue that under heavy class imbalance, the receiver-operating-characteristic (ROC) curve produces a misleading picture of classifier performance: "ROC plots in the context of imbalanced datasets can be deceptive ... owing to an intuitive but wrong interpretation of specificity" (2015, p. 1). The mathematical mechanism is that the false-positive rate places true negatives in its denominator, so when negatives vastly outnumber positives, even a poorly performing classifier registers a low FPR by arithmetic alone. The precision-recall plot avoids this trap by replacing the false-positive rate with precision, which is simply the share of positive predictions that turn out to be correct. Because precision counts only what the model said yes to, the huge pool of true negatives that drags the false-positive rate toward zero plays no role

at all. Saito and Rehmsmeier add a second consequence: "the AUC(PRC) of random classifiers is ... $P/(P+N)$ " (2015, p. 7). In plain terms, the chance line on a precision-recall plot is not a fixed 0.5 but moves with the prevalence of the positive class. A random classifier on a heavily imbalanced dataset achieves an AUPRC equal to that prevalence rather than 0.5, so any AUPRC above the prevalence reflects real signal and one at or below it is no better than guessing.

| Unified Theory of Acceptance and Use of Technology (UTAUT)

Developed by Venkatesh et al. (2003), UTAUT provides the analytical structure for the meso level interrogation in this paper. Namely in the analysis of organizational viability. UTAUT synthesizes eight prior technology acceptance models, including the widely used Technology Acceptance Model (TAM) by Davis (1989), into four constructs that help evaluate different aspects effecting the adoption of a technology. Each construct is enriched below with case-relevant theory, serving the purpose of providing grounded analytical depth.

Performance Expectancy

Performance Expectancy is defined as the degree to which an individual believes that using the system will help them attain gains in their job performance (Venkatesh et al., 2003). Related to this, Jarrahi et al. (2022) propose hybrid intelligence as a framework for organizational settings in which AI and humans contribute complementary capabilities. AI handles pattern recognition across high-dimensional data that exceeds human cognitive capacity, while humans contribute contextual judgment, ethical reasoning, and the capacity to interpret meaning in situations algorithms cannot model. Jarrahi's claim is that neither intelligence is sufficient alone, and that the organizational design problem is constructing workflows that leverage both. The central insight is that the value an AI system delivers is therefore not a property of the system in isolation but emerges from the interaction between human and algorithmic intelligence. This hybrid approach is an accepted and proven approach within AI-implementation in Danish clinical decision-making, as outlined by Lauritzen et al. (2024), in the case of radiological imaging. Performance Expectancy in a hybrid system is a property of the human-machine team rather than of the machine alone.

Effort Expectancy

Effort Expectancy is defined as the degree of ease associated with the use of the system (Venkatesh et al., 2003). This section is partly explored through Zuboff (1985), who argue that information technology has two fundamentally different capacities. It can automate, replacing human judgment with algorithmic execution, or it can informate, making previously invisible patterns visible to human decision-makers and thereby strengthening their capacity for judgment. The distinction is not a spectrum but a design choice with divergent organisational consequences. Automating technologies deskill the workforce and concentrate authority in the

system, while informing technologies create what Zuboff terms "intellective skills", enabling workers to operate at a higher level of abstraction. The framework was developed in the context of 1980s industrial process control and does not account for the acute cognitive constraints under which contemporary informing systems operate, where the recipient of the information may not have the cognitive bandwidth to interpret it regardless of how legibly it is presented, this limitation defines the empirical scope at which the framework requires supplementing, not the validity of the underlying distinction.

Social Influence

Social Influence is defined as the degree to which an individual perceives that important others believe they should use the system, an effect heavily moderated by the voluntariness of use (Venkatesh et al., 2003). This area is supported by Parasuraman and Riley (1997), who establish that trust in automated systems is not a fixed attribute but a dynamic calibration, built through repeated interaction and adjusted based on the system's observed reliability. The calibration is asymmetric. A single conspicuous failure erodes trust faster than multiple correct predictions build it. Three pathologies of miscalibrated trust are identified: disuse (underutilizing the system because trust is too low), misuse (over-relying on the system because trust is too high), and abuse (deploying the system in contexts for which it was not designed). Related to this, Cvach (2012) describes alarm fatigue as a condition in which clinicians, overwhelmed by the volume of clinically irrelevant alerts, develop systematic desensitization, delaying response to or entirely ignoring alarms that may be clinically significant. Alarm fatigue is not a failure of individual attention but a predictable system-level response to a signal-to-noise ratio that exceeds human processing capacity.

Facilitating Conditions

Facilitating Conditions are defined as the degree to which an individual believes that an organizational and technical infrastructure exists to support the use of the system. Venkatesh et al. (2003) establish that the influence of Facilitating Conditions on usage behavior is heavily moderated by experience, becoming increasingly critical as users mature with the technology. Facilitating Conditions in UTAUT are populated empirically rather than theoretically. They consist of the available infrastructure, governance frameworks, training resources, and budget envelopes within which the system must operate. Holden and Karsh (2010) extend the technology acceptance literature specifically into healthcare, arguing that standard constructs require adjustment for the unique constraints of clinical work. This requires considering not only basic IT infrastructure, but the budget envelopes (total cost of ownership and labor allocation) required to sustain the system. Additionally, the high stakes and stringent regulatory requirements of clinical work necessitate formal governance frameworks that explicitly define legal liability and multi-stakeholder accountability. The construct is therefore the most context-dependent of the four.

Methodology

| Philosophy of Science

The paper works from a Critical Realist position, paired with a pragmatic approach to method (Sayer, 1992). The choice stems from what the research question requires rather than from a prior philosophical commitment.

Sepsis is treated as something real that unfolds whether or not anyone detects it, and whose physiological signs exist independently of the devices used to record them. A strict positivist stance (Popper, 1959) cannot host the organizational half of the question, because hospital implementation is not a measurable property of the world but a contingent outcome of workflows, regulations, and trust. An interpretivist stance (Weber, 1949) cannot host the technical half, because reducing sepsis to a clinical interpretation loses the biological reality and technical findings in the paper. Critical Realism allows both layers in view, treating the underlying mechanisms, the events they produce, and the data those events leave behind as different but related parts of one reality.

Method follows from this commitment rather than the other way around. The paper uses quantitative machine learning to ask whether a wearable-proxy signal can capture deterioration, and UTAUT (Venkatesh et al., 2003) to interrogate the nature of how a human can act on that signal. The shared philosophical frame binds the two halves into one coherent investigation, while the methodological pragmatism (Saunders, Lewis & Thornhill, 2019) reflects what a desk-based analysis can credibly attempt. The conclusions and recommendation on implementation are therefore conditional, a reasoned argument about which conditions need to be considered for proposed system to deliver value.

| Research Design

To answer the research question, this paper employs a dual-movement analytical strategy, a structure necessitated by its Critical Realist foundation (Sayer, 1992). Because the biological deterioration of sepsis and the clinical treatment of sepsis represent two interacting but distinct strata of reality, the methodology must independently interrogate both the mathematical limits of biological prediction and the social realities of clinical implementation.

The Technical Interrogation

The first analytical movement is quantitative and empirical. Utilizing the MIMIC-IV critical care database (Johnson et al., 2023), this phase constructs a proxy for wearable-derived data to train and evaluate a stacked

machine learning ensemble against a rule-based clinical baseline. The objective of this movement is to establish the underlying empirical reality: can delimited wearable-capturable vital signs capture the non-linear trajectory drifts of sepsis (Sturmberg, 2021), and how do the perditions compare to current clinical standard?

The Organizational Translation

The second analytical movement shifts from the empirical to the social, translating the mathematical outputs of the first movement into organizational and economic realities. This phase utilizes the Unified Theory of Acceptance and Use of Technology (UTAUT) to evaluate whether a hospital could viably implement the system (Venkatesh et al., 2003). However, following Holden and Karsh's (2010) critique that generic technology acceptance models are insufficient for high-stakes, multi-stakeholder clinical environments, the framework is heavily contextualized using relevant theory. The statistical outputs generated in the analytical part are actively used to contextualize the UTAUT constructs.

Ultimately, this dual movement ensures that the paper does not treat machine learning as an isolated technological artifact. Instead, the algorithm's statistical performance acts as the direct context for the socio-technical and economic evaluation, addressing the literature gap between technological capabilities and organizational translation identified in the background section.

| Use of AI

This paper's utilization of GenAI was restricted to the following areas: to refine the written prose, namely sharpening clarity, flow, and sentence structure, while preserving the author's content, argumentation, and direction. Further, it was utilized to assist with the development and refinement of code across the data pre-processing, modelling, and visualization pipeline, including syntax corrections, debugging, optimization, and the rendering of figures.

| Ethical Considerations

This study relies on the secondary analysis of retrospective, de-identified health records sourced from the MIMIC-IV critical care database (Johnson et al., 2023). Because there was no primary data collection or direct interaction with patients, the research poses no direct risk to human subjects. Access to the dataset was securely provisioned via Google BigQuery under the strict terms of the PhysioNet Credentialed Data Use Agreement (Johnson et al., 2023). Furthermore, methodological compliance was ensured through the author's mandatory completion of the CITI human-subjects research training and subsequent institutional approval prior to any data extraction (Appendix 1). Ethical questions raised by what the paper proposes, rather than what it does, are taken up in the organizational analysis and the discussion.

| Data & Preprocessing

To execute the empirical interrogation outlined in the first analytical movement, this section details the systematic transformation of raw clinical records into a viable modeling foundation. It outlines the extraction, cleansing, and target-mapping of the MIMIC-IV database required to construct the wearable-proxy dataset that drives the machine learning pipeline.

Collection and Description

This analysis was developed, using the Medical Information Mart for Intensive Care (MIMIC-IV), a publicly accessible critical care database maintained by the MIT Laboratory for Computational Physiology (Johnson et al., 2023). MIMIC-IV contains de-identified health records for 546,028 adult hospitalizations at Beth Israel Deaconess Medical Center in Boston, Massachusetts, spanning the period from 2008 to 2022. The database is organized into three modules: [1] A Hospital module covering admissions, laboratory results, prescriptions and diagnoses, [2] an ICU module covering high-frequency bedside charted observations, inputs and outputs [3] and a Derived module of severity scores, treatment flags, and aggregated first-day summaries built on top of the other two.

Figure 1 - MIMIC-IV Module Breakdown

Module	Tables	Columns	Features
Hospital (<u>hosp</u>)	19	130	82
ICU (<u>icu</u>)	9	64	27
Derived (<u>derived</u>)	19	103	60
Severity scores	7	29	16
Treatments / organ support	4	16	9
Derived labs	4	27	18
First-day summaries	4	31	25
Total (schema-wide)	47	297	154

Figure 1 shows MIMIC-IV’s schema overview across its three 3 modules.

MIMIC-IV was selected because its’ scale allows a sepsis cohort large enough for robust machine learning, its’ resolution is sufficient to reconstruct datasets with lower sampling frequency than current clinical practices, and its documentation is mature enough to apply well-established derivations such as the Sepsis-3 suspicion-of-infection target, without having to redefine and argue their validity from scratch.

Mapping Target Variable

The target variable is Sepsis-3 follows the Third International Consensus definition (Singer et al., 2016), which characterizes sepsis as a life-threatening organ dysfunction caused by a dysregulated host response to infection and operationalizes onset as the concurrent presence of (a) a suspicion of infection and (b) an acute rise of at least two points in the Sequential Organ Failure Assessment score (SOFA, a six-organ-system clinical severity index). Two alternative labelling approaches are common in the sepsis-prediction literature and were considered for this study. Retrospective identification using ICD codes (International Classification of Diseases) is straightforward to implement, but ICD codes carry no onset timestamp (Rhee et al., 2017), so they do not meet the temporal-resolution requirement of a time-series prediction task. SIRS-based real-time labelling is

temporally anchored and is the basis of earlier prediction models such as Barton et al. (2019). In 2016 consensus however changed, away from SIRS towards the stricter Sepsis-3 definition, which grounds the use of Sepsis-3.

The labels are generated in-project. The reference SQL implementation of Sepsis-3 is maintained by the MIT Laboratory for Computational Physiology in the open-source mimic-code repository (Johnson, Stone, Celi & Pollard, 2018; github.com/MIT-LCP/mimic-code), and the three dependent scripts (`suspicion_of_infection.sql`, `sofa.sql`, `sepsis3.sql`) were executed in sequence against MIMIC-IV within the project's Google BigQuery workspace, in accordance with data owners terms of use. This produced a Boolean `sepsis3` flag and an onset timestamp `sepsis3_time` for every row in the dataset (Each row represents a patient's observations for the given monitoring window, anchored by the `charttime` variable indicating last monitoring). With `sepsis3_time` established, the horizon flags were produced using a simple Python script. The output materialized 3 Boolean flags per row "`is_sepsis_6h`", "`is_sepsis_12h`", and "`is_sepsis_stay`", indicating whether the patient represented in the row suffered sepsis onset within the respective time horizon after last monitoring.

The above calculation of the ground-truth sepsis labels necessitated the use of an extensive number of features from different dataset modules and was thereby not subject to later data delimitation.

Cohort Construction

First, the cohort is restricted to ICU stays, because the ICU module is the only part of MIMIC-IV which reliably contain a suitable vital sign sampling frequency needed to compose higher resolution datasets for wearable-proxy testing.

To keep the analysis aligned with the wearables proxy delimitation, the features in the ICU module were restricted to the six vital signs current wearables technology is capable of reliable measuring along with three context features.

Figure 2 - Raw Features

Feature	Type	Description
heart_rate	Integer (bpm)	Heart rate from bedside monitor.
resp_rate	Integer (breaths/min)	Respiratory rate.
spo2	Integer (%)	Blood oxygen level
temperature	Float (°C)	Skin/body temperature.

Feature	Type	Description
sbp	Integer (mmHg)	Systolic blood pressure
dbp	Integer (mmHg)	Diastolic blood pressure.
age	Integer (years)	Patient age at admission.
weight_kg	Float (kg)	Admission weight.
time_since_admission	Float (hours)	Hours elapsed between ICU intime and each measurement row.

Figure 2 shows the raw features extracted from the MIMIC-IV dataset.

Exploratory Data Analysis

This exploratory data analysis evaluates the cohort's summary statistics, data resolution, and base-feature correlations. These findings empirically inform our subsequent preprocessing decisions, feature engineering, and non-linear model selection.

Summary Statistics

Summary statistics on the extracted cohort (Figure 3) show 1,929,876 vital-sign measurement rows across the six vitals. Mean values for heart rate, respiratory rate and SpO₂ sit within clinically normal ranges, reflecting that ICU patients spend most monitored hours in supported stability, with deterioration occurring as episodic tail events (a desirable property for a model intended to detect the transition from stability to deterioration, since both classes must be well-represented in training). Two main observations are made. First, extreme values in the Min and Max columns (heart rate up to 280 bpm, SpO₂ as low as 20%, temperature down to 25.2 °C, diastolic BP up to 200 mmHg) lie outside plausible physiological ranges and point to sensor artefacts or charting errors that must be filtered. Second, temperature shows an unusually narrow relative spread (CV $\approx$ 1.1%, against 17–28% for the other vitals), concentrating its clinically informative signal in a small number of tail readings; this forecast limited predictive ability of the raw vital, and motivates future feature engineering to focus on abnormalities such as fever and hypothermia values.

Figure 3 - Vital-sign Summary Statistics

Feature	Count	Mean	Std Dev	Min	25th Pct	Median	75th Pct	Max
Heart Rate (bpm)	1,929,876	82.9	17.7	10	70	82	96	280

Feature	Count	Mean	Std Dev	Min	25th Pct	Median	75th Pct	Max
Respiratory Rate (br/min)	1,929,876	19.1	5.3	1	16	18	22	100
SpO ₂ (%)	1,929,876	96.5	2.9	20	95	97	98	100
Temperature (°C)	1,929,876	36.9	0.4	25.2	36.7	36.9	37.1	41.4
Systolic BP (mmHg)	1,929,876	122.5	21.3	40	109	122	135	299
Diastolic BP (mmHg)	1,929,876	66.9	15.5	20	58	67	76	200

Figure 3 shows the summary statistics for the six vital signs on the extracted cohort.

Data Resolution Evaluation

Figure 4 - Vital-Sign Sampling Resolution

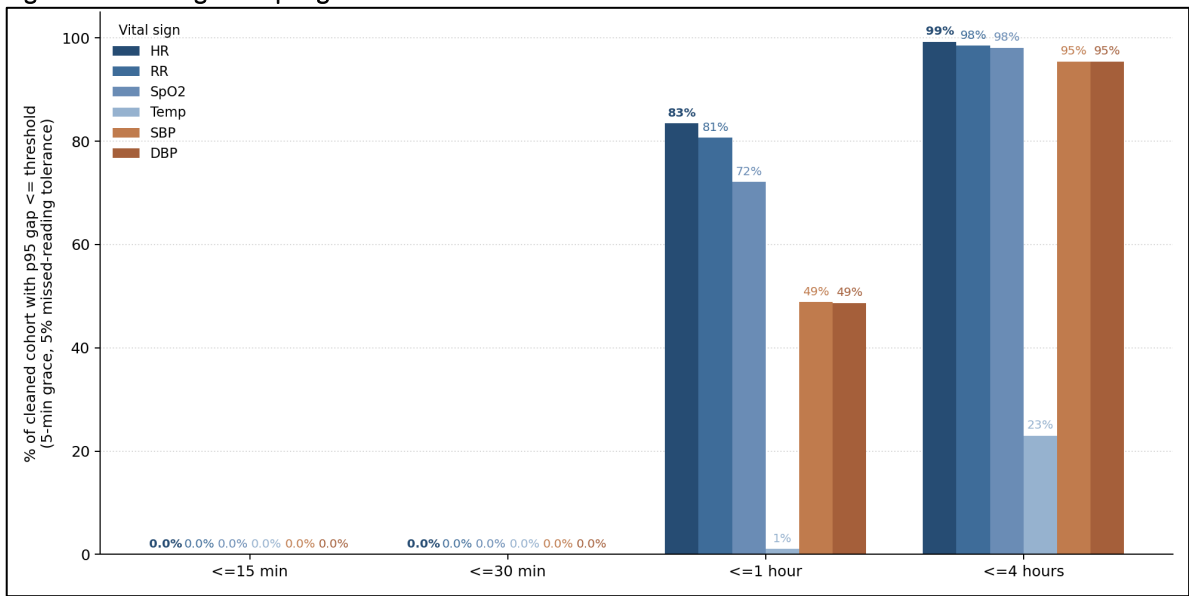


Figure 4 shows the percentage of the constructed MIMIC-IV cohort for which respective vital signs is sampled at or better than a given resolution.

Examination of sampling frequency available from the MIMIC-IV dataset show, that no vital for any patient is available at a 15 or 30-minute resolution, regrettably ruling out the construction of true-data based datasets in these lower resolutions. Fortunately, the 1-hour resolution shows promise, in regard to constructing a dataset with a higher sampling frequency than current ward standard. Here, the vitals split into three regimes, with the continuous autonomic signals well supported, blood pressure half grounded, and temperature isolated at the bottom as the bottleneck.

Base-Feature Evaluation

The pairwise correlations show that all raw vital signs carry weak linear signal against the sepsis labels, with the strongest being temperature ($r = +0.061$), followed by heart rate ($r = +0.029$). The vital signs themselves are moderately correlated in physiologically expected ways, with systolic and diastolic blood pressure sharing $r = +0.50$ and heart rate and respiratory rate sharing $r = +0.28$, but no pair approaches alarming correlation, which would implicate unnecessary noise. Weak univariate linear correlations imply that a simple linear model will struggle to discriminate cases, while the absence of multicollinearity means no feature needs to be dropped on correlation grounds alone.

Figure 5 - Correlation Matrix of Vital Signs & Context Features

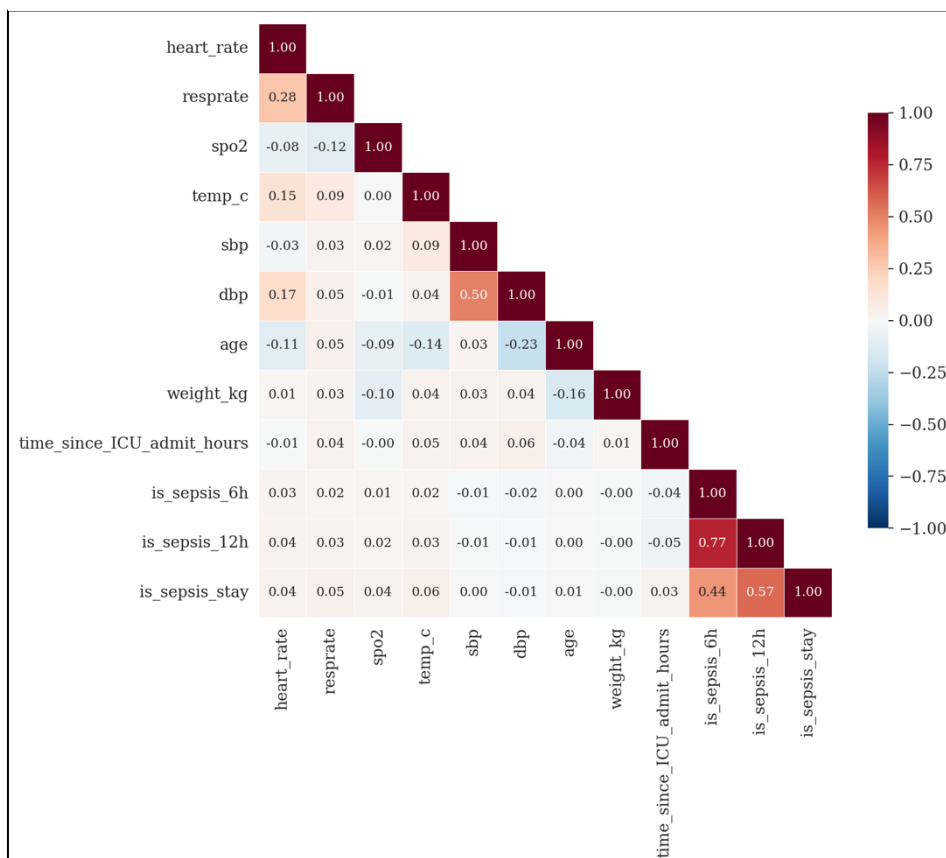


Figure 5 shows the pairwise pearson correlations among the six vital signs, context features, and three sepsis labels. Blue = positive correlation, red = negative, grey = near zero.

Figure 6 - Vital Sign Distributions, Sepsis vs. Non-Sepsis

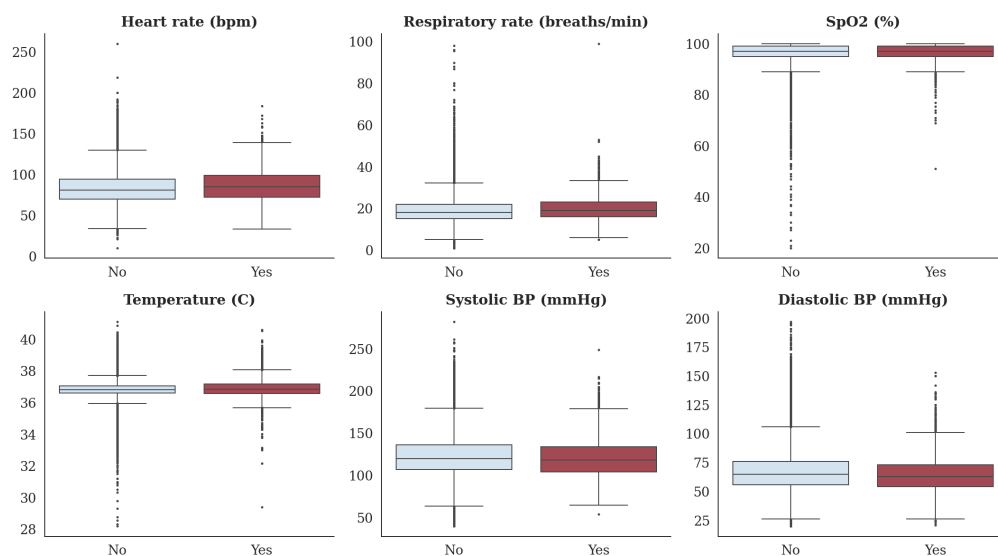


Figure 6 shows the distribution of each vital sign grouped by the 6-hour sepsis label (blue = non-sepsis, red = sepsis within 6 hours).

Median shifts between sepsis and non-sepsis rows are small for every vital sign individually. Heart rate and respiratory rate sit visibly higher in the sepsis group, and systolic and diastolic blood pressure sit slightly lower, with all six shifts pointing in the physiologically expected direction. The small univariate separations confirm the implication from Figure 5, that no single vital sign can carry the prediction, but the directionally consistent shifts across all six features suggests that a signal exists in distributed rather than localized form. The modeling problem is therefore framed as a multivariate pattern-recognition task rather than a threshold seeking task.

Resampling Mechanics

The findings in the Data Resolution Evaluation let to the construction of three datasets at different temporal resolutions to empirically test whether higher sampling resolution improves sepsis prediction. Dataset A is the 4-hour grid, constructed by forward-filling the last valid reading onto each grid point, mimicking the carry-forward logic of current manual nursing rounds. Datasets B and C are the 1-hour and 15-minute grids, both constructed by averaging readings within each bucket and linearly interpolating short gaps, which is the mean-and-interpolate rule appropriate to a resolution the cohort supports only partially.

Dataset A serves as the baseline of current ward practice. Dataset B is primary for testing whether higher sampling frequency improved predictive performance, because the continuous autonomic signals carry mostly real readings with only temperature heavily interpolated. Dataset C is fabricated resolution across the board, reported as an upper-bound extrapolation, risking the drowning out of true signal, at the benefit of more “synthetic” datapoints.

Figure 7 - Row-level Sepsis Prevalence

Dataset	ls_sepsis_6h	ls_sepsis_12h	ls_sepsis_stay
A	2.45%	3.51%	13.32%
B	1.70%	2.86%	13.32%
C	0.52%	0.87%	13.32%

Cleansing

MIMIC-IV is real hospital documentation, which means variable measurement frequencies, sensor artefacts, missing readings and heterogeneous patient populations. This necessitates a cleansing of the data, to minimize noise and optimize the data for ML. Three cleansing steps were applied.

[1] The cohort was restricted to ICU stays between 24 and 2 000 hours. Stays shorter than 24 hours do not provide enough data to compute reliable rolling-window features, while stays longer than 2 000 hours (≈ 83 days) are clear outliers whose sheer length distorts cohort statistics. [2] Physiologically impossible readings were removed, erring on the side of retaining plausible outliers rather than discarding genuine extreme readings that might themselves signal clinical deterioration. [3] Patients missing all readings for any of the six core vital signs were excluded, to avoid systematic missingness patterns that would otherwise confound learning.

The combined effect of these steps is shown in Figure 8. The cascade reduces the starting population from 94,458 ICU stays to a final modelling cohort of 37,285 patients (a retention rate of 39.5 %), with the bulk of the reduction driven by the length-of-stay window rather than by artefact removal.

Figure 8 - Cleansing Cascade

Cleansing / exclusion criterion	Dropped	Remaining
Drop non-ICU stays	451,570 patients	94,458 patients
Exclude patients under 18 (adults only)	0 patients	94,458 patients
Keep only ICU stays between 24 and 2 000 hours	56,972 patients	37,486 patients
Drop patients with zero chartevents rows in the window	23 patients	37,463 patients
Remove physiologically impossible readings	12,744 readings	37,463 patients
Drop patients missing one or more of the six core vitals entirely	178 patients	37,285 patients

Figure 8 shows the data-cleansing cascade from raw MIMIC-IV to the final modelling cohort (37,285 patients).

Figure 9 - Data Snippet

stay_id	charttime	age	weight_kg	time_since_admission	heart_rate	resp_rate	spo2	temperature	sbp	dbp	is_sepsis_stay	is_sepsis_12h	is_sepsis_6h
30004242	2180-11-19 01:00	75	77.6	0.13	73.5	21.0	97.0	36.72	107	90	1	0	0
30004242	2180-11-19 02:00	75	77.6	1.13	60.0	19.0	96.5	36.77	92	74	1	0	0
30004242	2180-11-19 03:00	75	77.6	2.13	46.0	16.0	96.0	36.82	132	47	1	0	0
30004242	2180-11-19 04:00	75	77.6	3.13	46.0	18.0	95.0	36.87	145	52	1	1	0
30004242	2180-11-19 05:00	75	77.6	4.13	48.0	17.0	94.0	36.91	125	58	1	1	0
30004242	2180-11-19 06:00	75	77.6	5.13	46.0	15.0	98.0	36.96	135	45	1	1	0
30004242	2180-11-19 07:00	75	77.6	6.13	49.0	18.0	98.0	37.01	133	48	1	1	0
30004242	2180-11-19 08:00	75	77.6	7.13	50.0	19.0	97.0	37.06	141	52	1	1	0
30004242	2180-11-19 09:00	75	77.6	8.13	47.0	18.0	97.0	36.96	117	50	1	1	0
30004242	2180-11-19 10:00	75	77.6	9.13	41.0	13.0	96.0	36.86	125	43	1	1	1
30004242	2180-11-19 11:00	75	77.6	10.13	46.0	16.0	97.0	36.76	137	47	1	1	1
30004242	2180-11-19 12:00	75	77.6	11.13	47.0	17.0	97.0	36.67	137	47	1	1	1
33003456	2180-11-19 02:00	30	80.0	0.60	47.0	18.0	97.0	36.96	117	50	0	0	0
33003456	2180-11-19 03:00	30	80.0	1.60	47.3	17.5	96.0	37.00	137	47	0	0	0

Figure 9 shows a fictive snippet of the dataset before feature engineering (as sharing true raw data from MIMIC is not allowed). This is an example from dataset B (1-hour resolution), where every row is a patient hour, with corresponding features and target flags. Stay_id is the unique stay identifier, and charttime is the time of latest vital-sign measurements.

| Modeling

The prediction task is a supervised binary classification problem on imbalanced clinical time-series data. Each patient-hour is labelled with whether sepsis will be diagnosed within the next six hours (or twelve), and the

model is trained to predict that label from the preceding feature history. Two structural features of the data shape the approach. The positive class is rare (under two percent of patient-hours at the six-hour horizon), which pushes toward methods that handle imbalance architecturally. Considering the nature and constraints of the application, the following section outlines this papers' attempt to build a suitable machine learning pipeline.

Establishing a Baseline and Exploring Architecture Capabilities

To create a reference point the choice was made to first established what a non-intelligent system would deliver. The scikit-learn dummy classifier with the 'prior' strategy, which always predicts the majority class, defines the chance. Four learning algorithms were then run as baselines, each chosen to test a specific architectural claim. Logistic Regression tests linear separability between the six wearable-proxy vitals and the sepsis label. A single Decision Tree probes non-linear univariate partitioning and exposed the variance problem Breiman (1996) identifies as the precondition for ensemble learning. Random Forest tests whether bagging that unstable learner reduces variance enough to lift signal above noise. XGBoost tests whether sequential boosting (Friedman, 2001) adds anything beyond bagging on this signal.

Standard-of-Care Comparator

To anchor the model's empirical performance against current Danish clinical practice, this paper computes a TOKS proxy. As explained in the background TOKS is the track-and-trigger scoring system used across Danish hospital wards today.

Figure 10 - TOKS Stages

TOKS-value	3 — ALARM	2 — UNSTABLE	1 — WARNING	0 — NORMAL
Respiratory rate (RR)	RF $\leq 7^*$, 25–29, 30–35, $\geq 36^*$	RF 21–24	RF 8–11	RF 12–20
Saturation (SpO ₂)	SpO ₂ 80–91, $\leq 79^*$	SpO ₂ 92–93	SpO ₂ 94–95	SpO ₂ ≥ 96
Supplemental oxygen	+2 added to the SpO ₂ score if patient is on supplemental oxygen			
Systolic blood pressure	SBP 80–89, $\leq 79^*$, ≥ 220	SBP 90–99	SBP 100–109	SBP 110–219
Heart rate (pulse)	HR $\leq 39^*$, 130–140, $\geq 141^*$	HR 110–129	HR 40–49 or 90–109	HR 50–89
Temperature	T 32–34.9, ≤ 31.9 °C*	T 39–40, ≥ 40.1 °C	T 35–35.9, 38–38.9 °C	T 36–37.9 °C
Consciousness (GCS)	GCS 9–14, $\leq 8^*$	—	—	GCS 15

The proxy mirrors the methodology of Nemati et al. (2018), who benchmarked the similar NEWS score, also using the MIMIC ICU dataset. The only adjustments are the substitution of the standard NEWS scoring family with the Region Nordjylland Danish TOKS 2.1 variant (Region Nordjylland, 2025). Concretely, each (patient, hour) row in the cohort is scored against the seven TOKS 2.1 variable thresholds, producing a continuous

TOKS_total ranging 0 to 18. The score is evaluated against is_sepsis_6h and is_sepsis_12h directly, with no clinical-judgment elements layered on top, as this variable is not feasible to estimate.

In machine-learning terms, the proxy is a scoring classifier, sometimes called a risk score in clinical-prediction terminology, namely it outputs a continuous TOKS_total value at each row rather than a binary sepsis prediction. To convert the score into a sepsis flag, an operating-point threshold must be picked. This paper is however not looking to find the optimal TOKS threshold, rather it will evaluate overall performance across all thresholds, using AURPRC.

Feature Engineering

The limited predictive value of individual raw vital signs and context features uncovered in the exploratory data analysis inspire this feature engineering section. In its essence, this section seeks to show the rationale, and nature of the features engineered for downstream model use. Feature engineering was largely a result of literature review, where features from different studies were examined and adopted if they showed high performance, without harmfully overlapping with existing features, and to a lesser degree some more explorative attempt to proxy non-available features proven to perform well in full feature studies (Allgöwer & Burri, 1967; Calvert et al., 2016; Churpek et al., 2016; Fleuren et al., 2020; Henry et al., 2015; Lauritsen et al., 2020; Mao et al., 2018; Moor et al., 2021; Morrill et al., 2020; Nemati et al., 2018; Reyna et al., 2020; Royal College of Physicians, 2017; Shashikumar et al., 2017; Singer et al., 2016; Subbe et al., 2001).

Clinical Composite Indices

Clinical-domain composites combine multiple raw vitals into single risk scalars by reproducing the bedside scoring systems (partial NEWS2, partial qSOFA, shock index) developed in the clinical literature with the exact purpose of detecting this sort of deterioration (Singer et al., 2016; Royal College of Physicians, 2017; Seymour et al., 2016). Their inclusion was a natural starting point given a wealth of retrospective validation showing composite scores outperform any single vital for early detection. The contribution here is not invention but selection, namely identifying which validated composites translate cleanly to a wearable-proxy feature set.

Figure 11 - Clinical Composite Features

Feature	Formula	In plain words
Shock Index	HR / SBP	Fast heart rate, weak blood pressure
Mean Arterial Pressure	$(SBP + 2 * DBP) / 3$	Average blood pressure
Modified Shock Index	HR / MAP	Stricter shock index
Pulse Pressure	$SBP - DBP$	Blood pressure spread
Rate-Pressure Product	$HR * SBP$	How hard the heart pumps
Automated NEWS Score	see NEWS2 calculation in Appendix 2	Light version of mainstream warning score

Partial qSOFA	see qSOFA calculation in Appendix 2	Light version of mainstream high-risk sepsis flag (quick Sequential Organ Failure Assessment)
Acute scoring deltas	score(t) - score(t-1), applied to NEWS and qSOFA	Sudden score change

Temporal Dynamics and Kinematic Features

Temporal dynamics convert each patient's vital-sign stream from disconnected snapshots into a trajectory through slopes, rolling variability, and drift from per-patient baselines. The approach follows a strand of sepsis prediction literature where trajectory features outperform point-in-time vitals, most prominently Morrill et al. (2020). Base-feature evaluation reinforced the choice, where many positive cases sat within ranges still labelled normal at any given instant, ignorant to development trends.

Figure 12 - Temporal Dynamics and Kinematic Features

Feature	Calculation	In plain words
Expanding minimum	minimum of x from start to now	Lowest seen so far
Expanding maximum	maximum of x from start to now	Highest seen so far
Expanding mean	mean of x from start to now	Average so far
Expanding standard deviation	SD of x from start to now	Spread so far
Rolling 4-hour SD	SD over trailing 4-hour window	Recent four-hour spread
EWMA (3-hour span)	exponentially weighted mean, 3-hour span	Smoothed recent average
4-hour slope	$x(t) - x(t-4)$	Change over four hours
Lagged difference (1 h)	$x(t) - x(t-1)$	Change in last hour
Lagged ratio (1 h)	$x(t) / x(t-1)$	Ratio to last hour
Acceleration	$\Delta(t) - \Delta(t-1)$	Change of change
CUSUM positive drift	running sum of $\max(0, x - \text{patient baseline})$	Drift above usual baseline
CUSUM negative drift	running sum of $\max(0, \text{patient baseline} - x)$	Drift below usual baseline

Biological Interaction Features

Biological interaction features combine two or more raw signals into ratios, differences, and products to proxy biological constructs that have shown predictive value in full-feature sepsis studies but are not observable from this paper's wearable-proxy data limitation (Ahmad et al., 2009; Casserly et al., 2015). The contribution here is exploratory, namely asking whether wearable-feasible signal combinations can approximate constructs the wearable cannot measure directly.

Figure 13 - Biological interaction features

Feature	Formula	In plain words
Cardiorespiratory Distress Index	$RR * (100 - SpO_2)$	fast breathing and low oxygen
Fever-Driven Tachycardia	$HR * \max(\text{Temp} - 37, 0)$	fast pulse with fever
Generalised Perfusion Adequacy	$HR / (SBP * SpO_2 / 100)$	heart works hard for oxygen

Cardiorespiratory Coupling	4-hour rolling Pearson correlation of HR and RR	heart and lungs in sync
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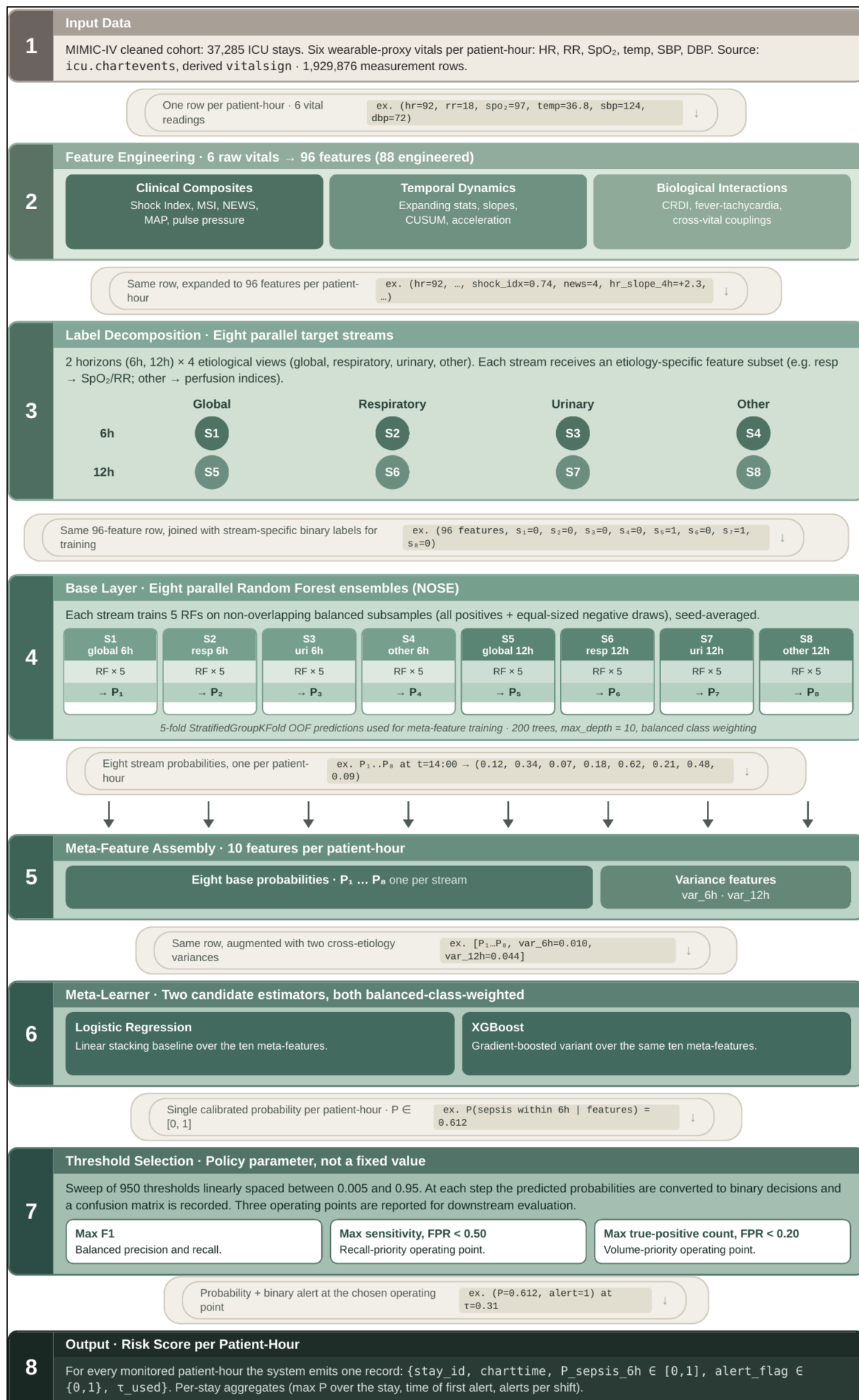
Preventing Look-Ahead Bias

Before detailing the modeling, it is nice to establish how the pipeline prevents look-ahead bias. Three main forms of data leakage are addressed explicitly. [1] Training data is restricted to observations charted at or before the Sepsis-3 onset timestamp, with everything thereafter excluded, so the model is trained to predict the window leading up to onset rather than to detect post-onset physiology after the fact. [2] The train-test split is executed entirely at the patient-level rather than the row-level, preventing the meta-learner from memorizing patient-specific baseline trajectories. [3] Every engineered feature introduced in the preceding section are computed causally. By strictly restricting calculations to past values relative to the prediction time, these controls define an absolute look-ahead boundary that the model must honor.

Stacked Ensemble

The stacked architecture is licensed by Wolpert's (1992) stacked generalization, which combines multiple base learners by training a meta-learner to deduce and correct their systematic biases. Two design moves follow. First, the base layer must contain learners with structurally different errors so that there is bias to deduce; this is achieved by partitioning the prediction problem into eight streams along sepsis etiology and prediction horizon, which gives each base learner a different hypothesis space. Second, the base learners must each be variance-controlled, because a meta-learner over noisy base outputs simply propagates the noise; this is achieved by making each stream a Random Forest (Breiman, 2001) with NOSE-bagged training.

Stacked Ensemble Pipeline Diagram



Level 3 - Label Decomposition

The stacked ensemble breaks sepsis prediction into eight parallel sub-tasks rather than training one all-purpose classifier. The decomposition is licensed by Wolpert's (1992) stacking framework, which requires diversity in base-learner errors for the meta-learner to have something to correct, and is justified clinically because sepsis presents differently depending on its source: lung infections shift oxygen saturation and respiratory rate first, urinary infections shift temperature and heart rate first, abdominal and bloodstream infections shift blood pressure first.

The cohort is divided under a fixed priority order, ICD-10 J13–J18, J20–J22, J69 (ICD-9 480–488, 507) routes to respiratory, ICD-10 N10–N12, N30, N39 (ICD-9 590, 595, 599) routes to urinary, and any remaining sepsis stay routes to other, with the resulting assignment written into the training and test datasets as the binary labels `target_resp`, `target_uri`, and `target_other`.

Level 4 - NOSE

Only a small fraction of patients carries a positive sepsis label. A single learner trained on the raw distribution would therefore likely collapse to the trivial prediction. Three standard remedies are rejected: class weighting reweights the loss but does not address the underlying scarcity of positives; synthetic oversampling (Chawla et al., 2002) interpolates between minority neighbors and produces vital-sign combinations the body cannot generate; single-pass under-sampling discards most of the negative pool. The Stacked Ensemble instead adopts a Negative-Oversampled Subset Ensemble (NOSE), in the family Liu et al. (2009) introduce as EasyEnsemble. Positive stays are paired with five disjoint negative subsets of equal size, a Random Forest is trained on each, and the outputs are averaged per stream. The result expands negative coverage, reduces variance through bagging (Breiman, 1996), and preserves real physiological observations throughout.

Base-Learner Hyperparameter Selection

Each Random Forest's regularization and ensemble parameters were selected empirically rather than left at scikit-learn defaults. Those defaults are calibrated for general-purpose performance across a wide range of problems and are agnostic to the architectural choices made upstream (NOSE balancing, stay-level cross-validation) and to the asymmetric cost structure of clinical alerting that determines what counts as a useful operating point. A one-factor-at-a-time grid search therefore varied each parameter across a defensible range with all others held fixed, scored each candidate on AUROC, AUPRC, and F1 on fold 0 of the 4-hour cohort, and retained the value at which the metric curves saturate. The final configuration is `depth 8`, `min_samples_leaf 35`, `max_features sqrt`, `n_estimators 150`, `class_weight None`, `bootstrap True`, with five NOSE subsets per stream. Each subsequent claim about ensemble behavior and meta-learner bias correction therefore rests on base learners measured against the data they are actually applied to, rather than on borrowed defaults.

Etiology-specific feature subsets

What prevents the eight ensembles from collapsing into near-identical classifiers is a constraint on the feature mix each one sees. Respiratory streams weight SpO₂ and respiratory-rate derivatives, urinary streams weight temperature and tachycardia markers, other streams (abdominal, skin, surgical-site, bloodstream) weight blood-pressure-derived indices, and global streams retain the full feature panel as a generalist comparator. The architecture mirrors Breiman's (2001) strength-and-correlation ideas but applied at the ensemble level: feature partitioning forces the eight learners into different hypothesis spaces, which reduces inter-stream correlation.

Level 5 - Meta-feature assembly

At Level 5, the eight base probabilities are combined with two cross-etiology variance features (representing the degree to which the base-learner predictions disagree) into a ten-dimensional input to the meta-learner. The structural form follows Wolpert's (1992) stacked generalization, where the meta-learner is trained on the predictions of the base learners as features and on the true outcomes as targets. Variance carries diagnostic meaning in this setup: converging probabilities across the four etiological lenses signal an unambiguous case, diverging probabilities flag base-layer uncertainty, and uncertainty itself is predictive of borderline outcomes.

Level 6 - Meta-learners

At Level 6, two candidate estimators (Logistic Regression and an XGBoost) are tasked to predict sepsis onset from the results delivered by the base-learners and the variance scores. In this, they consider the sum of the prediction scores from the eight base-learners, as well as the degree to which they disagree.

Level 7 - Threshold selection

Like the base-learner, the trained meta-learners, do not output a binary decision, but a continuous probability score denoted as a float between 0 and 1 representing its confidence that the patient will develop sepsis within the prediction horizon. Converting that score into an actionable alert requires a decision threshold, the cutoff above which the probability triggers a positive prediction. Thresholds were not fixed at the conventional 0.50 but were picked from a sweep of 950 thresholds linearly spaced between 0.005 and 0.95. At each step the predicted probabilities were converted to binary decisions and the resulting confusion matrix recorded. Three example thresholds were then selected: the threshold maximizing F1 (balanced precision and recall), the threshold maximizing TPR subject to FPR < 0.50, and the threshold maximizing true-positive count subject to FPR < 0.20.

Evaluating Estimator Performance

This section reports the results of all compiled estimators. These estimators were evaluated across the three resolution datasets as outlined in the resampling mechanics section. All estimators were trained on 75% of their dataset cohort. The trained estimators were then tested against the remaining 25%, by their ability to predict the “is_sepsis_6h” or “is_sepsis_12h” targets respectively. Results are represented using performance metrics which are known to be suitable for imbalanced prediction problems.

- **AUCROC:** The Area Under the Receiver Operating Characteristics curve, weighs an estimator’s ability to rightly predict sepsis cases (true positive rate (TPR) aka. Recall aka. Sensitivity), and how frequent the estimator wrongly predicts a healthy patient will get sepsis (false positive rate (FPR)).
- **AUCPRC:** The Area Under the Precision Recall Curve, weighs an estimator’s ability to rightly predict sepsis cases (Recall), and proportion of predicted sepsis cases which turn out to actually be sepsis (Precision).
- **F1-score:** Harmonic mean of precision and recall. This metric takes both false positive and false negatives into account.

Figure 14 - Model Results for Prediction of Sepsis Onset Within Next 6 Hours (4-hour resolution)

Version	Estimator	Threshold	AUPRC	PPV	AUROC	Recall	F1	TP %	FN %	FP %	TN %
Stacked Ensemble	XGBoost (meta).	0.7299	0.0663	0.0855	0.7289	0.2434	0.1265	6.10%	1.89%	6.49%	91.02%
Stacked Ensemble	XGBoost (meta)	0.3934	0.0663	0.0401	0.7289	0.815	0.0764	2.03%	46.00%	48.66%	48.84%
Stacked Ensemble	XGBoost (meta)	0.5945	0.0663	0.0611	0.7289	0.5075	0.1091	1.26%	1.23%	19.42%	78.09%
Stacked Ensemble	Logistic Regression (meta)	0.717	0.0653	0.0816	0.7275	0.2566	0.1238	6.40%	1.85%	7.20%	90.31%
Stacked Ensemble	Logistic Regression (meta)	0.3525	0.0653	0.0399	0.7275	0.8119	0.0761	2.02%	47.00%	48.69%	48.82%
Stacked Ensemble	Logistic Regression (meta)	0.5955	0.0653	0.0608	0.7275	0.5037	0.1084	1.26%	1.24%	19.40%	78.11%
Engineered Features	XGBoost	0.5	0.0650	0.0541	0.7246	0.5963	0.0992	1.49%	1.01%	25.98%	71.53%
Engineered Features	XGBoost	0.3	0.0650	0.0391	0.7246	0.8258	0.0746	2.06%	0.43%	50.64%	46.87%
Engineered Features	Random Forest	0.5	0.0602	0.0575	0.7175	0.4767	0.1026	1.19%	1.30%	19.49%	78.02%
Engineered Features	Random Forest	0.3	0.0602	0.0431	0.7175	0.7487	0.0815	1.87%	0.63%	41.46%	56.05%
Engineered Features	Logistic Regression	0.5	0.0553	0.0454	0.7027	0.6871	0.0851	1.71%	0.78%	36.02%	61.49%
Engineered Features	Logistic Regression	0.3	0.0553	0.0293	0.7027	0.9268	0.0568	2.31%	0.18%	76.58%	20.93%
Raw Vitals	XGBoost	0.5	0.0399	0.0776	0.6205	0.0103	0.0181	0.03%	2.49%	0.31%	97.17%
Rule-Based Baseline	TOKS Proxy	3	0.0387	0.0316	0.6424	0.829	0.061	2.09%	0.43%	63.99%	33.49%
Rule-Based Baseline	TOKS Proxy	5	0.0387	0.0387	0.6424	0.673	0.073	1.70%	0.82%	42.17%	55.31%
Rule-Based Baseline	TOKS Proxy	7	0.0387	0.0428	0.6424	0.317	0.075	0.80%	1.72%	17.85%	79.63%
Rule-Based Baseline	TOKS Proxy	10	0.0387	0.0528	0.6424	0.063	0.057	0.16%	2.36%	2.84%	94.64%
Raw Vitals	Random Forest	0.5	0.0387	0.0392	0.6148	0.3982	0.0713	1.00%	1.52%	24.59%	72.89%
Raw Vitals	Logistic Regression	0.5	0.0311	0.0304	0.5665	0.5437	0.0576	1.37%	1.15%	43.61%	53.87%
Dummy Classifier	Stratified Dummy	0.5	0.0252	0.0253	0.5001	0.0246	0.025	0.06%	2.46%	2.38%	95.10%

Figure 14 shows the performance of all estimators, trained on dataset A, predicting whether patients would get sepsis within 6 hours. Tables for all resolutions and for prediction window 12h can be found in Appendix 3

| Key Performance Metrics

Comparing models trained at different thresholds and across different pipeline configurations requires a metric that is independent of any single decision-threshold. The standard choice is the Area Under the Receiver Operating Characteristic Curve (AUROC). Under severe class imbalance, however, AUROC produces misleading evaluations. Because the false positive rate places true negatives in its denominator, the large pool of non-

sepsis instances arithmetically compresses FPR, flattering estimators that have learned to identify healthy patients rather than deteriorating ones.

Figure 15 - AUROC vs. AUPRC

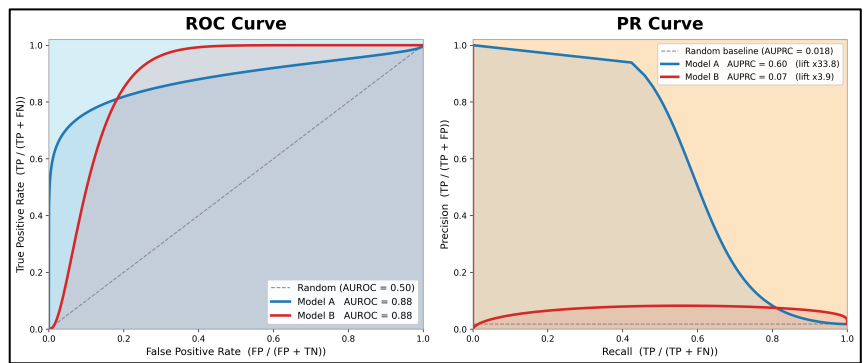


Figure 15 shows the performance of the same two fictive estimators.

The examples in Figure 15 illustrates this issue. Looking at the ROC diagram, both example model A and B have an AUROC of 0.88, even though they have very different developments. Placing these same two example estimators on a precision recall diagram produces a much more useful comparison, which does not care about how good model B is at predicting the majority negative class, but only how good it is at predicting the minority positive sepsis class. Furthermore, AUPRC's random-classifier baseline equals the positive-class prevalence, so subtracting prevalence from a reported AUPRC gives a simple read of the estimators lift over chance.

| Overall Performance

Figure 16 - Estimator Performance

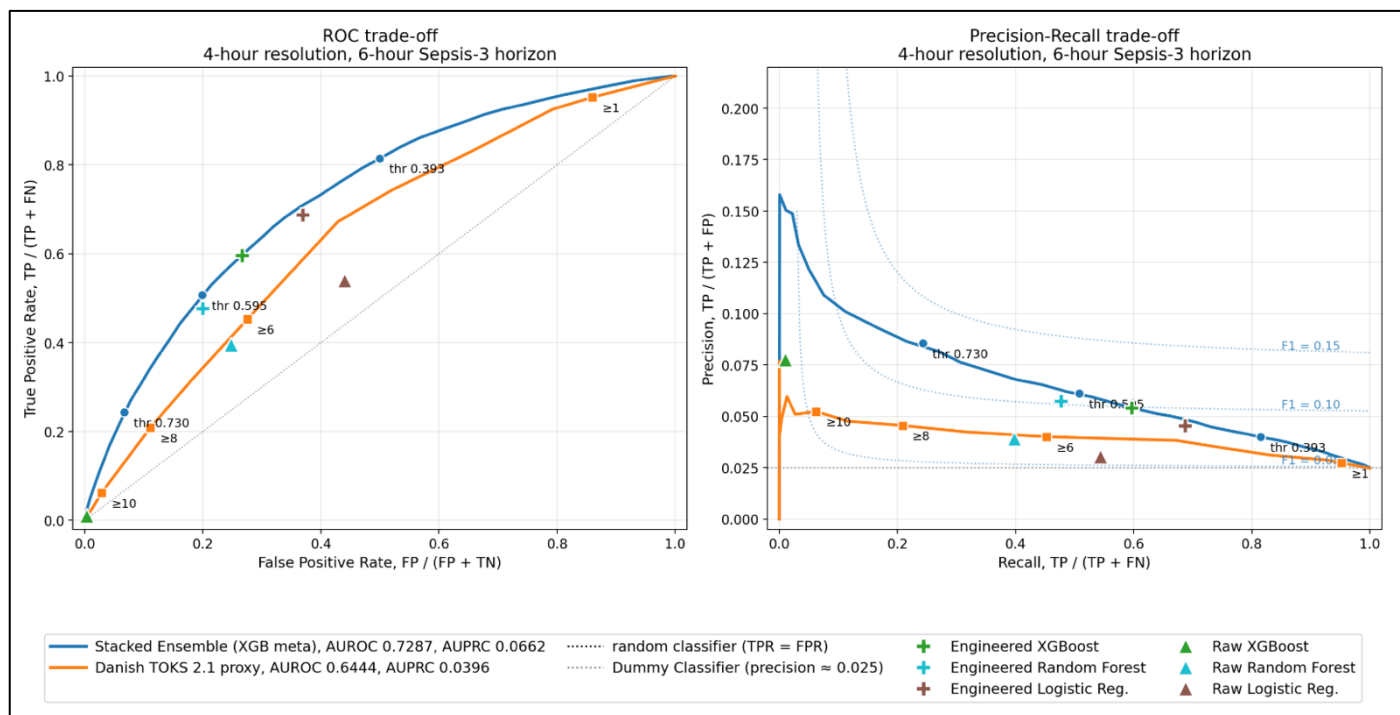


Figure 16 shows the point performance of all estimators, versus the Stacked Ensemble estimator developed in this paper and a TOKS proxy roughly approximating current clinical reality. To fairly compare TOKS to the other estimators, the 4-hour resolution was chosen, as this represents current clinical sampling frequency.

With an AUPRC of 0.0252, the random classifier tells us that 2.5% of patient-rows had positive `is_sepsis_6h` flag and establishes the chance of 1x. From here, every tier raises the lift above random, namely 1.23x to 1.58x across the raw-vitals tier, 1.57x at the TOKS proxy, 2.19x to 2.58x across the engineered single-model estimators, and 2.59x to 2.63x at the stacked ensemble tier. Each tier represents a measurable improvement over the one below it. The stacked ensembles lift above chance, shows a 67% jump in sepsis-specific discrimination, compared to TOKS proxy.

This tells us; [1] Estimators trained on the raw vitals perform worse or comparably to the rule-based TOKS proxy, indicating that using machine learning does not improve performance on its own. [2] Estimators trained on the engineered features vastly outperform the same models trained on raw vitals. [3] The Stacked Ensemble outperforms every other estimator, indicating that the sum of architectural and tuning decisions made improved predictive performance.

If we assume Danish nurses follow the documented TOKS protocol strictly, the results also implies that machine learning fed with just six wearable-capturable vitals can predict sepsis onset better than the current clinical standard.

| Effect of Feature Engineering

The outperformance of estimators trained on engineered feature datasets, show that feature engineering as a whole contributed predictive value. To understand each feature's marginal predictive signal independent of any specific estimator's tuning or interaction effects, univariate lift is calculated by measuring AUPRC using only that single feature as the predictor, then calculating lift. Anything above 1x shows above random predictive ability.

In panel (a) in Figure 17 below, one feature family dominates. CUSUM drift features, which communicate deviation from a patient-specific running mean, at lifts between 1.36x and 1.55x across every channel. Additionally the illustration shows that most features carry predictive ability (even on their own).

Lag ratios, single-hour differences, and accelerations cluster near or below 1x. Indicating random or negative independent predictive value.

Panel b sheds light on a few methodological wins for this paper. Three features entirely novel to this paper, Fever-Driven Tachycardia at 1.26x, Cardiorespiratory Distress Index at 1.17x, and Generalized Perfusion Adequacy at 1.15x, outperform established features like Partial NEWS (1.11x) and Partial qSOFA (1.03x), with Fever-Driven Tachycardia also clearing the established Modified Shock Index at 1.22x. Fever-Driven Tachycardia

additionally sits below the engineered set's median pairwise correlation (Appendix 4), indicating significant unique discriminative value rather than redundant signal already captured by other features.

Figure 17 - Feature Importance Heatmaps



Figure 17 shows univariate AUPRC lift per feature in the stacked ensemble pipeline (1-hour resolution, 6-hour target). Panel (a) arranges kinematic features as transformation x vital-sign channel; panel (b) lists composites, biological interactions, and contextual features.

In essence. Trajectory features, not instantaneous values, carry the highest predictive value.

| Effect of Base Layer Specialization

Figure 18 - Performance of Specialized Base Learners Against own Target

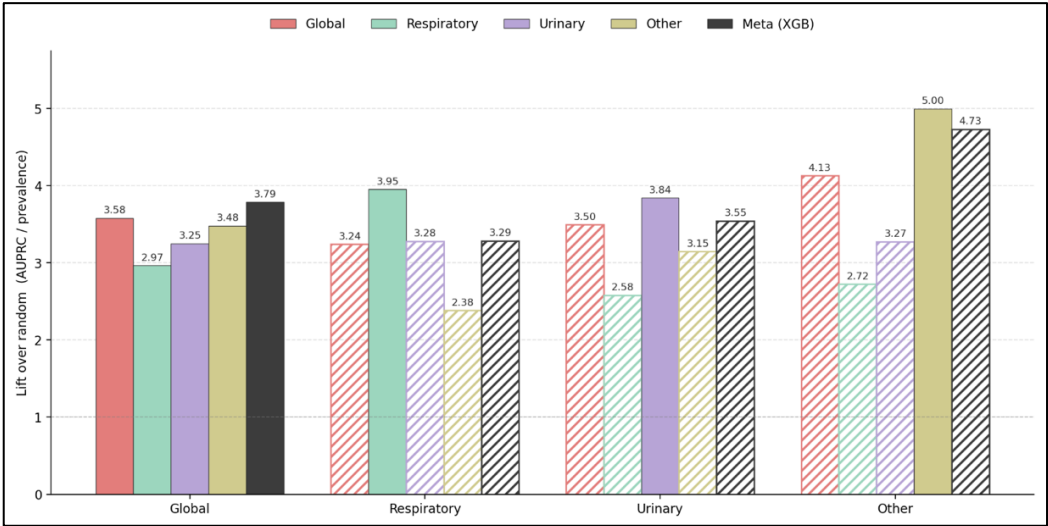


Figure 18 shows the four ensembles and the meta learner's ability to predict sepsis onset within 6 hours. They are trained on the 1-hour resolution dataset and scored against held-out test with four binary labels, namely the global `is_sepsis_6h` flag and three etiology-specific flags. The etiology labels are strict subsets of the global label, so prevalence falls from 2.45% globally to 0.84% for respiratory, 0.36% for urinary, and 1.24% for other. Each AUPRC is divided by the relevant label's prevalence to yield a lift above random.

Each specialized base learner predicts its own etiology more accurately than others, confirming that the base learners develop the subgroup-specific discrimination the stacking design intended. Each stream contributes etiology-specific signal a single classifier cannot recover.

Ultimately, we see that the XGB meta learner, build on the outputs of the specialists, is the best overall performing estimator, beating the global estimator in every category.

| Effect of Threshold Setting

Figure 19 - Confusion Matrices with Differing Thresholds

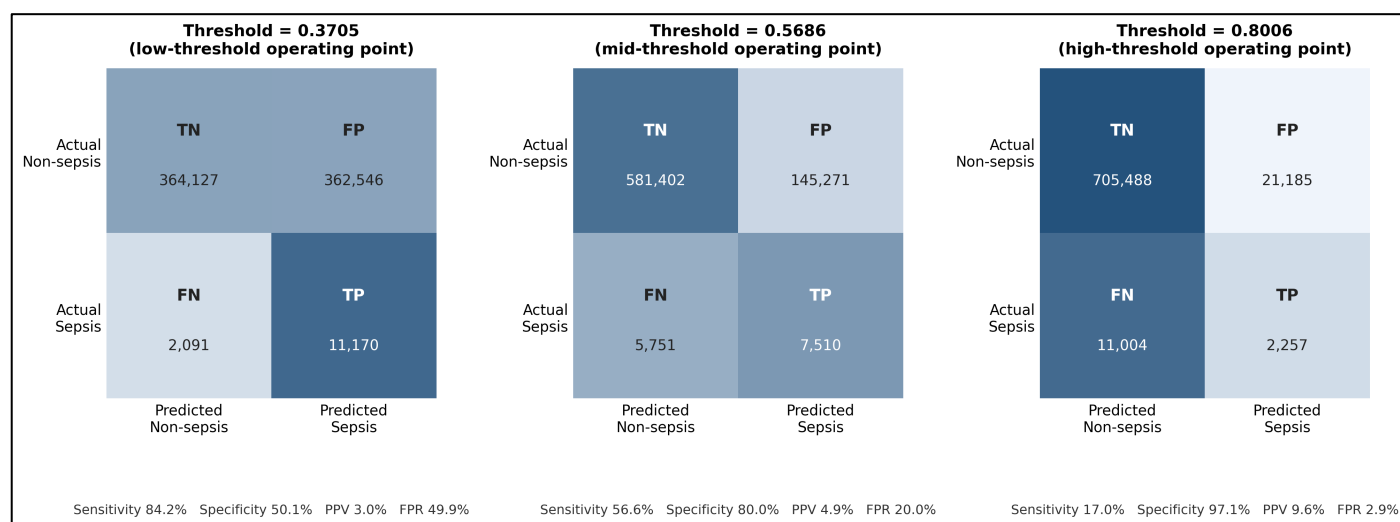


Figure 19 shows the confusion matrices for different decision thresholds. The stacked ensemble with the XGBoost meta-learner trained on dataset B (1 hour resolution), predicting sepsis within 6 hours was chosen as a representative example. All estimators examined showed similar reaction to changing of the decision-threshold.

The confusion matrices for the stacked ensemble decompose its discrimination ceiling into three concrete operating choices. At threshold 0.3705 the model captures 11,170 of 13,261 sepsis cases (84.2% TPR) at the cost of 362,546 false alerts. Raising the threshold to 0.5686 retains 7,510 cases (56.6% TPR) while cutting false alarms by roughly 60%. At 0.8006 the model crosses into specificity-first territory, flagging only 2,257 cases against 21,185 false alarms but missing 11,004 true cases.

The key insight is that decision-threshold selection, dramatically shifts the nature of an estimator's output, weighing the maximization of true sepsis onset prediction against the tolerance for false alarms.

A last observation is that false alerts per captured case do not fall proportionally with the threshold. At threshold 0.3705 the model produces 32.5 false alerts per true positive. At 0.5686 the ratio drops to 19.3, and at 0.8006 it falls to 9.4. This is likely a result of the severe class imbalance. False positives saturate the low-confidence region of the probability distribution, so only the most confident predictions lift precision meaningfully, and tightening the cutoff therefore buys disproportionate alarm-fatigue relief.

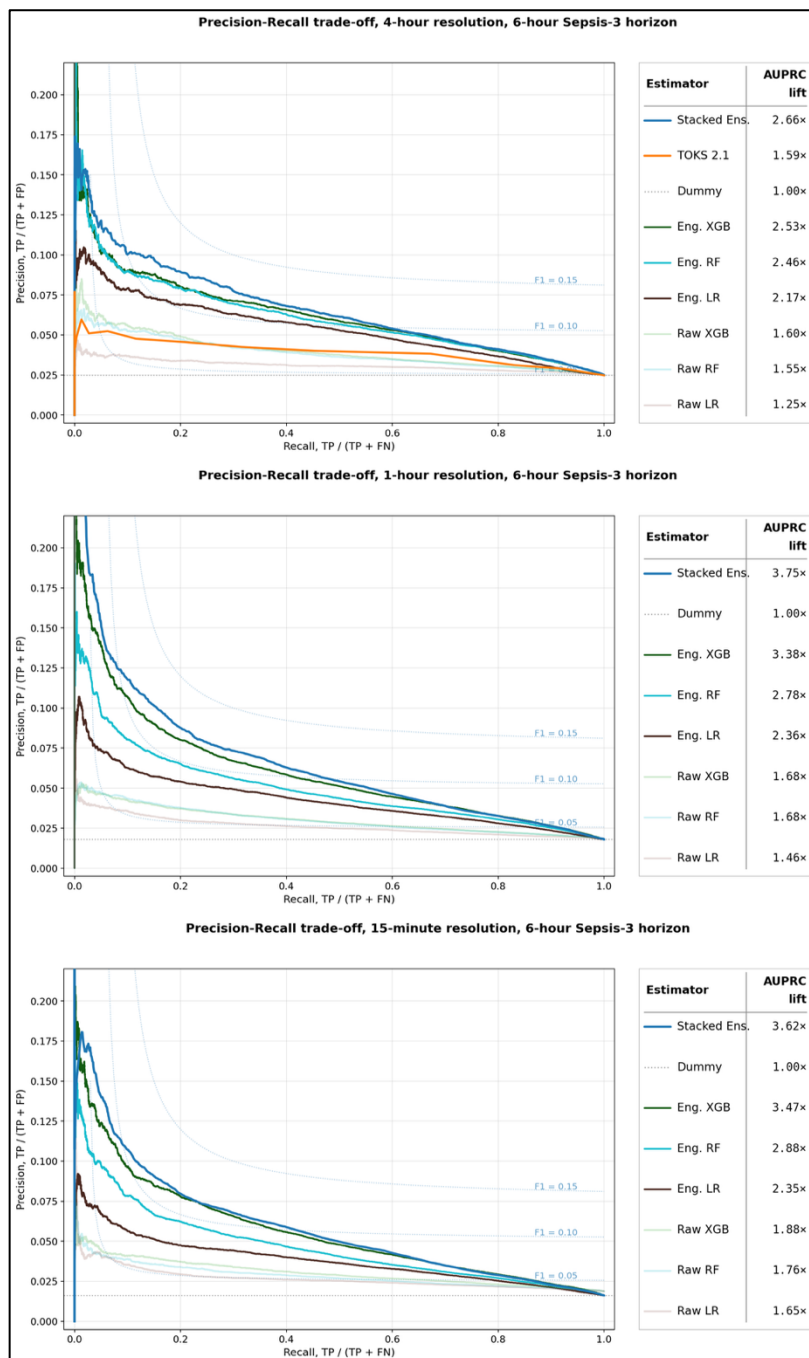
| Effect of Higher Data Resolution

Sampling resolution shapes the estimators' view of patient trajectory more than any other pre-processing decision. The comparison between the 4-hour resolution that mirrors current ward rounds and the 1-hour cadence that represents the highest reliable resolution in MIMIC-IV carries the load-bearing finding. Every single estimator improves performance significantly when trained on the 1-hour resolution dataset compared to the 4-hour dataset. The stacked ensemble climbs from an AUPRC lift of 2.66x at 4 hours to 3.75x at 1 hour, a 41% jump in sepsis-specific discrimination achieved by nothing other than sampling more often.

Pushing further, the experimental 15-minute resolution, all estimators improve, except for the stacked ensemble, which actually falls significantly from 3.75x AUROC lift to only 3.62x. While it is still the best estimator, this development is interesting. One explanation could be in the synthetic nature of most datapoints in this resolution. As explained in the exploratory data analysis the MIMIC dataset's highest resolution

with high real data availability is 1 hour. As explained in the resampling mechanics section, the 15-minute resolution dataset was therefore produced by resampling and forward-filling sparser real measurements, which consequently also means that as resolution rises, the proportion of informationally novel rows falls. The AUPRC lift improves anyway for weak learners because they were data-starved (underfit) at 1h and any extra

Figure 20 - PR at Different Data Resolutions



rows, even correlated ones, sharpen their decision boundary. The stacked ensemble may however already have been near the ceiling of the available information at 1h, so the additional rows mostly inject auto-correlated noise into the meta-learner rather than new signal.

Another plausible explanation might lay in the logic of the stacked meta-learner structure that powers the stacked ensemble. A stacked meta-learner is trained on out-of-fold base-model predictions, and its statistical efficiency depends on the de-correlation of base learners' residuals. When the input data becomes more auto-correlated, the residuals of the base learners become more correlated with one another because they are all making the same kind of mistake on the same forward-filled stretches (now making up $\frac{3}{4}$ of the training data). The covariance of the base-prediction matrix that the meta-learner regresses on inflates, the conditioning of that regression worsens, and the meta-learner overfits to error patterns that do not generalize. Wolpert (1992) and Breiman (1996) both emphasized that stacking gains shrink as base learners' errors become correlated, which is exactly the regime that the forward-filling approach tested might produce. If this is the corporate, the undesired performance might be solved through the use of true data, rather than the experimental forward-filled data constructed in the 15-minute resolution dataset. This may even hold promise of true significant performance enhancement, as it expands the number of informationally novel rows.

Organizational & Economic Viability

This section applies the four UTAUT constructs introduced in the theoretical framework (Venkatesh et al., 2003) to evaluate the conditions under which the proposed solution would be implementable on a Danish general ward. The solution is the deployment of wearable devices to capture vital signs, feed a machine learning model predicting sepsis-3 onset, and surface the results to clinical staff. The analysis works through Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions.

| Performance Expectancy

Following the reasoning from Jarrahi et al. (2022), performance expectancy on the ward cannot be assessed by looking at the algorithm in isolation, it must be evaluated as a property of the human-machine team that ultimately decides whether a flagged patient is escalated or dismissed.

The technical evidence presented in the data analysis chapter shows the machine (the ML prediction system) fulfils its half. The stacked ensemble outperforms the TOKS proxy on every measured metric at four-hour resolution. The performance lift traces directly to feature engineering, and the etiological specialization based

meta-learner structure, which allow for deeper understanding of complex interactive features. This edge cannot be replicated within current clinical standards, which rely on a rule-based decision support system. Therefore, results indicate that the machine contributes performance, the human cannot achieve without it.

While the machine contributes a unique edge in hard number prediction, it does not show perfect performance. Not all patients for which the machine will predict sepsis will actually suffer from sepsis and not all patients the machine will predict not to suffer from sepsis will avoid sepsis. It should therefore not unilaterally dictate admission of treatment nor dismissal of a patient, necessitating human involvement. The machine is still reliant on the human for enriching decision making with their unique contextual knowledge, not captured by the machine. The proposed machine can for instance detect a statistically anomalous heart rate trajectory, but it cannot currently distinguish sepsis-driven tachycardia from anxiety, post-operative pain, or a missed dose of a beta blocker. A black box paired only with a confidence score however reproduces the automation bias and disuse dynamic established in the theoretical framework (Parasuraman & Riley, 1997). Therefore, the human must be able to understand the underlying reasoning for recommendations surfaced by the machine, in order to evaluate whether the machine's recommendation is invalidated by a confounder-feature to which the machine is blind. Following the reasoning of Jarrahi et al. (2022), the effective human-machine translation will lead to the highest performance. Therefore, the systems performance expectancy, is contingent on showing ability to effectively translate reasoning to clinical staff.

Because Venkatesh et al. (2003) define Performance Expectancy at the individual level, the system's predictive accuracy must ultimately translate into tangible gains for the frontline clinician's daily workflow. Currently, Danish nurses perform periodic, manual vital-sign readings and documentation, a highly repetitive task that consumes significant time. Implementation of the proposed system would largely render this manual data collection redundant. Therefore, the primary performance gain for the individual nurse is twofold: it physically eliminates busy, repetitive labor, and reclaims their limited time so it can be reallocated toward more meaningful, high-impact efforts. Furthermore, the system empowers the nurse to directly improve patient outcomes.

| Effort Expectancy

Effort Expectancy (EE) captures the perceived ease of use associated with the system (Venkatesh et al., 2003). While the system reduces the physical effort of manual rounds, it risks replacing it with unsustainable cognitive load.

While the technical results suggests that the system would predict sepsis onset better than current clinical standard. If these predictions are deployed as loud, binary alarms dictating a prescribed clinical action, the

system falls into what Zuboff (1985) terms the "automating" trap. An automating approach attempts to substitute human judgment with algorithmic execution. If nurses trust and comply with this design, it leads to overreliance on the system, deskilling the workforce by removing them from the active decision-making process. Furthermore, while these alerts would statistically have fewer false alarms per true positive, overreliance and blind trust might result in nurses not applying their unique contextual knowledge to critically evaluate the situation, which would otherwise improve predictive ability (Jarrahi et al., 2022). Alternatively, an opaque predictor which is still wrong in by far most cases might lead to the opposite of obedience, the clinical literature confirms that low-explainability black box systems risks triggering severe alarm fatigue (Cvach, 2012; Graham & Cvach, 2010).

Conversely, an unfiltered informing approach (Zuboff, 1985) is equally detrimental to Effort Expectancy. If the system attempts to avoid automation by communicating the full complexity of the estimator system displaying all 96 complex engineered features, and the minute details of how each influenced the prediction, it would be overwhelmingly difficult for a ward nurse to interpret and understand, especially during a busy shift. Processing this volume of raw probability data requires immense intellectual skill and would risk just replacing the physical labor of vital checks with severe cognitive load and could easily cause information overload (Zuboff, 1985).

To secure a positive Effort Expectancy, the user interface must actively navigate between the automating trap and the risk of over-informing.

| Social Influence

Social Influence (SI) reflects the degree to which individuals perceive that "important others" believe they should use the system (Venkatesh et al., 2003). In healthcare IT, "important others" include patients and their families (Holden & Karsh, 2010). Removing the manual TOKS round eliminates a recurring social touchpoint for patients. If patients experience this as a loss of human contact, nurses might feel social pressure to reject the technology. Hospital leadership might need to redirect parts of the reclaimed nursing time into more social-comfort related tasks, so that patients experience the change as something positive, and the social pressure on the nurse converts from potential resistance into endorsement.

Furthermore, while hospital leadership can mandate the physical use of the technology (exerting top-down SI), Venkatesh et al. (2003) establish that the power of SI is heavily moderated by Voluntariness. Leadership cannot mandate nurses' cognitive engagement. Mølgaard et al. (2023) document that Danish nurses already frequently exercise their clinical judgement by negotiating recommendations of warning systems. Trust in clinical automation is built through dynamic calibration and is highly asymmetric; a single false alarm erodes

trust much faster than correct predictions build it (Parasuraman & Riley, 1997). Given the high false alarm ratio, trust risks to quickly erode, leading to the pathology of disuse. This indicates, that if the system lacks explainability, nurses will exercise their functional voluntariness through behavioral workarounds, such as ignoring alerts, effectively neutralizing the administrative mandate. Critically, nurses might ignore the systems recommendations in the cases where the system's prediction differ from the normal protocols prediction, which are possibly the very cases where the system's ability to comprehend complex interactive feature gives it a predictive edge.

| Facilitating Conditions

Facilitating Conditions (FC) encompass the objective organizational, technical, and economic infrastructure that exists to support the use of the system (Venkatesh et al., 2003). In clinical IT, Facilitating Conditions differ from the standard UTAUT operationalization. Holden and Karsh (2010) argue that the construct must absorb the high-stakes, regulated, multi-stakeholder reality of healthcare rather than reducing simply to network access and helpdesk availability. The viability question for the proposed system therefore breaks into three conditions: whether the economic incentives to fund it are aligned, whether the technical infrastructure to integrate it exists, and whether the governance frameworks to operate it safely can be put in place.

Organizational Readiness and The Economic Budget Envelope

Given the command-and-control nature of hospital operations, where the voluntariness of use is limited, we are not only interested in the frontline personnel's acceptance. For potential pilot testing and long-term adoption, hospital leadership is the ultimate decision-maker. At the leadership level, the viability of the system extends to how implementation will affect overall organizational performance and resource allocation. While a specific capital budgeting exercise exploring implementation costs, the holistic internal rate of return (IRR), and the project payback period is beyond the scope of this paper, the Total Cost of Ownership (TCO), including wearable devices, continuous server uptime, and IT support staff, is substantial.

However, because Denmark operates on a tax-financed, single-payer model with capped regional budgets (Birk et al., 2024), hospitals largely internalize the downstream costs of the entire patient pathway. It is reasonable to assume that the earlier and better prediction provided by the system would lower patient admission times and the number of patients transferring to the intensive care unit, while simultaneously lowering patient mortality. As asserted in background, each ICU case and escalation is exceptionally costly, and the clinical literature demonstrates that each hour of delayed treatment increases sepsis mortality by 7.6% (Kumar et al., 2006). Preventing these late-stage escalations generates the argument for the upfront investment.

Weighing in the fact that Danish hospitals currently face a severe structural shortage of approximately 4,700 nurses (Birk et al., 2024), further bolsters the case, as saving nursing labor might save hospitals from paying expensive overtime, or hiring above market rate to fill shortfall.

Technical Infrastructure

A system computing sensitive patient data, directly responsible for patient wellbeing requires high reliability and data-security. This would likely necessitate investing in local compute infrastructure required to operate the model. The Danish Sundhedsdataloven and the GDPR govern where patient data may be processed, which favors on-premise or regional health IT inference over commercial public clouds. Continuous monitoring of every general ward bed implies continuous inference, which translates into round-the-clock server availability, failover capacity at clinical-critical uptime, and audit-ready access logging. The binding technical FC is therefore whether the regional IT organization can absorb the operational burden of running a clinical ML inference layer to the needed uptime and security standard.

IT Governance and Liability

On the governance side, the legal infrastructure surrounding the system is central. If the system is deployed as a hybrid intelligence tool (Jarrahi et al., 2022) and a nurse exercises clinical judgment to override a high-confidence ML alert, the question of professional liability arises if the patient subsequently deteriorates. Viability requires the hospital to formally absorb algorithmic outputs into its clinical governance protocol. Without a written protocol that specifies decision rights, documentation requirements, and liability boundaries between the algorithm, the nurse, and the institution, staff might reject the system to protect against liability.

Technical Outperformance vs. Reality of Hybrid Intelligence

The empirical results demonstrate that the stacked ensemble significantly outperforms the rule-based TOKS proxy, despite being limited to fewer raw features. This implies that the current limit on ward-level sepsis prediction is not a lack of underlying physiological signal, but rather the inability of manual tools to algorithmically transform that signal into a clinical-decision-support output. Beyond improved discrimination, higher data resolution naturally causes more frequent patient re-evaluations. This effectively front-runs current manual practice by a sampling frequency delta of up to 3.99 hours. Because each hour of delayed treatment increases sepsis mortality by 7.6% (Kumar et al., 2006), this algorithmic lead time would allow practitioners to administer life-saving interventions significantly earlier, lowering overall mortality.

However, arguing from a position of pure technological optimism ignores the realities of clinical care. While the model beats the TOKS mathematical screen, Danish nurses do not rely on vital signs alone; they employ

active clinical judgment based on tacit visual cues and holistic concern an algorithm cannot replicate. Therefore, the proposed system is not designed to operate autonomously. Instead, viewed through Jarrahi et al.'s (2022) theory of hybrid intelligence, the results suggest that the algorithm serves only to upgrade the non-human analytical component of the monitoring equation. Its clinical viability remains entirely dependent on the nurse's ability to interpret the system's output and apply the contextual and ethical judgment that the machine cannot provide.

Furthermore, while the etiological specialization of the base learners successfully reduced variance, deploying the algorithm requires confronting the severe risk of alarm fatigue. If the hospital attempts to manage this by deploying loud, binary escalation alarms, it falls into Zuboff's (1985) "automating" trap, attempting to substitute human judgment with algorithmic execution. Given the noise ratio, this would guarantee severe alarm fatigue, actively desensitizing nurses to the system. Parasuraman and Riley (1997) establish that trust in clinical automation is asymmetric and fragile; when users cannot understand a system's reasoning, they rapidly lose trust. Conversely, applying Zuboff's informing strategy by surfacing all 96 underlying features and probabilities to inform the nurse risks severe cognitive overload. If a clinician cannot intuitively grasp why the complex meta-learner is flagging a patient, the system risks triggering trust erosion and the behavioral pathologies of either disuse or blind obedience. Both outcomes undermine Jarrahi et al.'s (2022) vision of a hybrid intelligence team. Ultimately, if the cognitive load is too high, the system fails to secure a positive Effort Expectancy (Venkatesh et al., 2003), rendering the technology organizationally unviable regardless of its technical performance.

Methodological Considerations

The empirical results of the machine learning analysis inherently justify the methodological dualism of this paper. From a strict positivist perspective, the problem of sepsis prediction might appear "solved" by the technical data: the stacked ensemble model successfully captures more sepsis cases with better discrimination than the clinical proxy. However, as established in the preceding chapter, maximizing this mathematical sensitivity generates a high false-alarm ratio that fundamentally threatens the clinical workflow. Viewed through the lens of Critical Realism (Sayer, 1992), the biological mechanism of sepsis deterioration is an objective reality that the machine learning model successfully maps. Conversely, the treatment of sepsis and the implementation of the monitoring system are socially constructed events, highly contingent upon human workflows, trust, and cognitive load.

If this study had relied solely on the quantitative machine learning results, it would have concluded that the system was a resounding success. By introducing the UTAUT framework (Venkatesh et al., 2003), the analysis moves beyond treating the system as an isolated predictive tool. However, successfully applying this framework required confronting a broader theoretical limitation. As Holden and Karsh (2010) argue, generic technology acceptance models were originally built for office workers and often fail to capture the unique, high-stakes constraints of clinical work. This analysis validates their critique. Evaluating this system required heavily contextualizing UTAUT: replacing generic "ease of use" with Zuboff's cognitive load theories, and replacing standard IT infrastructure metrics with the economic reality of clinical Facilitating Conditions within a single-payer system. Ultimately, this demonstrates that researching healthcare IT requires bridging the empirical reality of the algorithm with the social reality of the ward.

Limitations & Future Research

Future research must primarily address the proxy boundaries and the desk-research-based organizational understanding this study relies on. Because the analysis utilized the MIMIC-IV dataset, it operates on the bridging assumption that commercial wearables deliver identical precision to invasive ICU telemetry. Future research must empirically validate wearable sensors on general wards to ensure real-world noise does not worsen the system's false-alarm ratio. Furthermore, the clinical proxy constructed as baseline is merely an automated rule-based screen that lacks the contextual and visual judgment applied by actual nurses. Future clinical validation must therefore evaluate the algorithm against true bedside nursing performance rather than a strict algorithmic proxy. Finally, the organizational analysis relies on desk research for UTAUT contextualization, lacking empirical beliefs elicitation from the target end-users. Future research should investigate these socio-technical factors through primary qualitative studies. Technically, algorithmic development could evaluate unsupervised clustering to group specialized base-learners, offering an alternative to the current reliance on retrospective clinical documentation for etiology-specific routing. Additionally, backward-ablation A/B testing is recommended for low-lift features to determine if their removal reduces computational load without omitting significant non-linear interactions.

Conclusion & Recommendations

This paper set out to investigate the case for using continuous wearable monitoring to predict sepsis onset for patient on Danish hospital wards.

The first analytical movement concludes that the technical bottleneck in ward-level sepsis detection is not a lack of underlying physiological data, but rather the inability of manual, rule-based systems to transform that data. By employing an etiology-specific stacked ensemble strictly constrained to wearable-proxy vital

signs, the model achieved a 67% improvement in sepsis-specific discrimination over current clinical standard proxy at a 4-hour resolution. Further, the higher sampling frequency enabled by wearables shows an additional 41% jump in sepsis specific discrimination at 1-hour resolution, indicating that increasing sampling frequency alone improves sepsis onset prediction ability. Coincidentally, the higher monitoring frequency would also result in earlier alarms, granting clinical personal additional lead time to initiate treatment sooner, which is critical given that each hour of delayed intervention increases sepsis mortality by 7.6%.

However, the second analytical movement, driven by a contextualized UTAUT framework (Venkatesh et al., 2003; Holden & Karsh, 2010), affirms that this predictive accuracy is entirely subordinate to the organizational and behavioral infrastructure governing its use. Given the mathematical realities of predicting rare deterioration events, the model retains a high false-positive rate. The analysis argues that treating this technology as a pure labor-substituting, automating decision making to combat the shortage of 4,700 Danish nurses is unviable, risks causing severe alarm fatigue (Cvach, 2012) and deskilling the workforce.

| Recommendations

To bridge this gap between predictive performance and implementation viability, this paper makes three strategic recommendations to guide potential implementation.

[1] To lower the cognitive effort expectancy and mitigate alarm fatigue, the algorithm's output must not be delivered as binary escalation alarms. Instead, it should be deployed through a 1–3-tiered risk-score interface that mirrors the existing TOKS 2.1 cognitive routines. Risk Score 1 should operate in an "informing" mode (Zuboff, 1985) by surfacing trajectory features for routine assessment, Score 3 as a semi-automating escalation alert, and Score 2 as a blended information prompt.

[2] The system must be designed for Hybrid Intelligence (Jarrahi et al., 2022). It must provide clinicians with the ability to interrogate which specific features are driving the risk score. This transparency builds the necessary trust and allows nurses to enrich the algorithmic assessment with their unique, contextual clinical knowledge.

[3] To navigate the complex social influences and liability concerns surrounding algorithmic overrides, hospitals must avoid a broad initial implementation. Deployment should begin on a single pilot ward to safely calibrate alert thresholds against real-world sensor noise, establish clear clinical governance protocols, and build frontline user trust before expanding hospital wide.

Ultimately, this paper finds a strong case for leveraging wearables and machine learning for sepsis prediction, while recognizing the need for future research to address the proxy limitations and desk-research based organizational understanding this paper relies on.

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Appendices

| Appendix 1 - Data or Specimens Only Research Training Completion Certificate

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM) COMPLETION REPORT - PART 1 OF 2 COURSEWORK TRANSCRIPT**

** Scores on this Transcript Report (Part 2) reflect the most current quiz completions, including quizzes on optional (supplemental) elements of the course. The Requirements Report (Part 1) lists the reported scores at the time all requirements for the course were met.

• Name: Nikolaj Stom Petersen (ID: 15258872)
• Institution Affiliation: Massachusetts Institute of Technology Affiliates (ID: 1912)
• Institution Email: nikolajstompetersen@gmail.com
• Institution Unit: Management

• Curriculum Group: Human Research
• Course Learner Group: Data or Specimens Only Research
• Stage: Stage 1 - Basic Course

• Record ID: 74864958
• Current Score*: 97

REQUIRED, ELECTIVE, AND SUPPLEMENTAL MODULES	MOST RECENT	SCORE
History and Ethics of Human Subjects Research (ID: 498)	24-Jan-2026	4/5 (80%)
Basic Institutional Review Board (IRB) Regulations and Review Process (ID: 2)	24-Jan-2026	5/5 (100%)
Belmont Report and Its Principles (ID: 1127)	24-Jan-2026	3/3 (100%)
Records-Based Research (ID: 5)	24-Jan-2026	4/4 (100%)
Genetic Research in Human Populations (ID: 6)	24-Jan-2026	5/5 (100%)
Populations in Research Requiring Additional Considerations and/or Protections (ID: 16680)	24-Jan-2026	5/5 (100%)
Research and HIPAA Privacy Protections (ID: 14)	24-Jan-2026	5/5 (100%)
Conflicts of Interest in Human Subjects Research (ID: 17464)	24-Jan-2026	5/5 (100%)
Massachusetts Institute of Technology (ID: 1290)	24-Jan-2026	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid Independent Learner.

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COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM) COMPLETION REPORT - PART 1 OF 2 COURSEWORK REQUIREMENTS*

* Scores on this Requirements Report (Part 1) reflect quiz completions at the time all requirements for the course were met. The Transcript Report (Part 2) lists more recent quiz scores, including those on optional (supplemental) course elements.

• Name: Nikolaj Stom Petersen (ID: 15258872)
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• Curriculum Group: Human Research
• Course Learner Group: Data or Specimens Only Research
• Stage: Stage 1 - Basic Course

• Record ID: 74864958
• Completion Date: 24-Jan-2026
• Expiration Date: 24-Jan-2029
• Minimum Passing: 90
• Reported Score*: 97

REQUIRED AND ELECTIVE MODULES ONLY	DATE COMPLETED	SCORE
Belmont Report and Its Principles (ID: 1127)	24-Jan-2026	3/3 (100%)
History and Ethics of Human Subjects Research (ID: 498)	24-Jan-2026	4/5 (80%)
Basic Institutional Review Board (IRB) Regulations and Review Process (ID: 2)	24-Jan-2026	5/5 (100%)
Records-Based Research (ID: 5)	24-Jan-2026	4/4 (100%)
Genetic Research in Human Populations (ID: 6)	24-Jan-2026	5/5 (100%)
Populations in Research Requiring Additional Considerations and/or Protections (ID: 16680)	24-Jan-2026	5/5 (100%)
Research and HIPAA Privacy Protections (ID: 14)	24-Jan-2026	5/5 (100%)
Conflicts of Interest in Human Subjects Research (ID: 17464)	24-Jan-2026	5/5 (100%)
Massachusetts Institute of Technology (ID: 1290)	24-Jan-2026	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid Independent Learner.

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| Appendix 2 - Calculation of NEWS2 Score & qSOFA

NEWS2 Score calculation
Sum of 5 per-vital sub-scores (each 0 to 3). Total ranges 0 to 15.
HR (bpm): 3 if <=40 or >=131; 2 if 111-130; 1 if 41-50 or 91-110; 0 if 51-90
RR (breaths/min): 3 if <=8 or >=25; 2 if 21-24; 1 if 9-11; 0 if 12-20
SpO2 (% , scale 1): 3 if <=91; 2 if 92-93; 1 if 94-95; 0 if >=96
Temperature (degC): 3 if <=35.0; 2 if >=39.1; 1 if 35.1-36.0 or 38.1-39.0; 0 if 36.1-38.0
SBP (mmHg): 3 if <=90 or >=220; 2 if 91-100; 1 if 101-110; 0 if 111-219
Source: Royal College of Physicians (2017); validation in Smith et al. (2013).
Partial qSOFA Score calculation
Sum of two binary points (each 0 or 1). Total ranges 0 to 2.
+1 if SBP < 100 mmHg
+1 if RR >= 22 breaths/min
Note: omits the GCS / altered-mentation criterion (not wearable-feasible).
Source: Singer et al. (2016), Sepsis-3 consensus definition.

I Appendix 3 - Model results

For is sepsis_6h

15-MINUTE RESOLUTION																
Version	Model	AUPRC	Precision (PPV)	AUROC	F0.5	F1 Mac	F2	FPR	FNP	TP	FN	FP	TN	Threshold	Recall	F1
Stacked Ensemble	XGBoost (meta)	0.0587	0.0858	0.7563	0.0955	0.5464	0.1451	0.0307	0.8245	8228	38659	87708	2764954	0.7966	0.1755	0.1152
Stacked Ensemble	XGBoost (meta)	0.0587	0.0269	0.7563	0.0333	0.3596	0.1191	0.4981	0.1631	39241	7646	1420996	1431666	0.3595	0.8369	0.0521
Stacked Ensemble	XGBoost (meta)	0.0587	0.0449	0.7563	0.0551	0.4846	0.1709	0.1994	0.4291	26770	20117	568794	2283868	0.5636	0.5709	0.0833
Stacked Ensemble	Logistic Regression (meta)	0.0585	0.0792	0.7507	0.0902	0.5438	0.1539	0.0384	0.7988	9436	37451	109634	2743028	0.7787	0.2012	0.1137
Stacked Ensemble	Logistic Regression (meta)	0.0585	0.0263	0.7507	0.0327	0.3587	0.1167	0.4988	0.1789	38497	8390	1422914	1429748	0.3246	0.8211	0.051
Stacked Ensemble	Logistic Regression (meta)	0.0585	0.0444	0.7507	0.0545	0.4842	0.1689	0.199	0.437	26399	20488	567761	2284901	0.5656	0.563	0.0824
Engineered Features (Single Model)	XGBoost	0.0577	0.0432	0.7515	0.053	0.4819	0.1648	0.2028	0.4432	26108	20779	578409	2274253	0.5	0.5568	0.0802
Engineered Features (Single Model)	XGBoost	0.0577	0.0297	0.7515	0.0367	0.3929	0.129	0.425	0.2095	37066	9821	1212474	1640188	0.3	0.7905	0.0572
Engineered Features (Single Model)	Random Forest	0.0501	0.0415	0.7392	0.0508	0.4828	0.1563	0.1933	0.4912	23857	23030	551560	2301102	0.5	0.5088	0.0767
Engineered Features (Single Model)	Random Forest	0.0501	0.029	0.7392	0.0359	0.3918	0.1262	0.426	0.2255	36316	10571	1215109	1637553	0.3	0.7745	0.0559
Engineered Features (Single Model)	Logistic Regression	0.039	0.0319	0.7155	0.0394	0.4347	0.1329	0.3172	0.3638	29828	17059	904881	1947781	0.5	0.6362	0.0608
Engineered Features (Single Model)	Logistic Regression	0.039	0.0197	0.7155	0.0245	0.2078	0.0907	0.7674	0.0634	43913	2974	2189052	663610	0.3	0.9366	0.0385
Raw Vitals (No Feat. Eng.)	XGBoost	0.0307	0.073	0.6269	0.019	0.4995	0.0059	0.0012	0.9952	36	7516	457	392806	0.5	0.0048	0.0089
Raw Vitals (No Feat. Eng.)	Random Forest	0.0301	0.0308	0.625	0.0378	0.4504	0.12	0.263	0.5651	3284	4268	103424	289839	0.5	0.4349	0.0575
Raw Vitals (No Feat. Eng.)	Logistic Regression	0.0269	0.0248	0.5942	0.0307	0.3872	0.1054	0.4241	0.4382	4243	3309	166796	226467	0.5	0.5618	0.0475
Dummy Classifier	Stratified Dummy	0.0189	0.0208	0.501	0.0207	0.501	0.0206	0.0185	0.9795	309	14794	14532	771867	0.5	0.0205	0.0206
1-HOUR RESOLUTION																
Version	Model	AUPRC	Precision (PPV)	AUROC	F0.5	F1 Mac	F2	FPR	FNP	TP	FN	FP	TN	Threshold	Recall	F1
Stacked Ensemble	Logistic Regression (meta)	0.0634	0.0819	0.7523	0.0938	0.5448	0.1664	0.0459	0.7757	2974	10287	33327	693346	0.7648	0.2243	0.12
Stacked Ensemble	Logistic Regression (meta)	0.0634	0.0293	0.7523	0.0363	0.3614	0.1283	0.499	0.1746	10946	2315	362581	364092	0.3246	0.8254	0.0566
Stacked Ensemble	Logistic Regression (meta)	0.0634	0.0491	0.7523	0.0601	0.488	0.1822	0.1991	0.4362	7477	5784	144677	581996	0.5766	0.5638	0.0904
Stacked Ensemble	XGBoost (meta)	0.0677	0.0963	0.7576	0.1054	0.5503	0.1475	0.0292	0.8298	2257	11004	21185	705488	0.8006	0.1702	0.123
Stacked Ensemble	XGBoost (meta)	0.0677	0.0299	0.7576	0.037	0.362	0.1309	0.4989	0.1577	11170	2091	362546	364127	0.3705	0.8423	0.0577
Stacked Ensemble	XGBoost (meta)	0.0677	0.0492	0.7576	0.0601	0.4878	0.1824	0.1999	0.4337	7510	5751	145271	581402	0.5686	0.5663	0.0905
Engineered Features (Single Model)	XGBoost	0.0625	0.0445	0.7583	0.0547	0.4708	0.173	0.2426	0.3804	8217	5044	176275	550398	0.5	0.6196	0.0831
Engineered Features (Single Model)	XGBoost	0.0625	0.0306	0.7583	0.0379	0.3685	0.1335	0.4857	0.1603	11135	2126	352977	373696	0.3	0.8397	0.059
Engineered Features (Single Model)	Random Forest	0.0553	0.0435	0.7411	0.0534	0.4764	0.1653	0.2203	0.4507	7284	5977	160065	566608	0.5	0.5493	0.0807
Engineered Features (Single Model)	Random Forest	0.0553	0.0303	0.7411	0.0376	0.371	0.1321	0.479	0.1789	10888	2373	348097	378576	0.3	0.8211	0.0585
Engineered Features (Single Model)	Logistic Regression	0.0432	0.0359	0.7176	0.0443	0.4375	0.1472	0.3195	0.3475	8653	4608	232137	494536	0.5	0.6525	0.0681
Engineered Features (Single Model)	Logistic Regression	0.0432	0.022	0.7176	0.0273	0.2189	0.1005	0.7537	0.0712	12317	944	547662	179011	0.3	0.9288	0.043
Raw Vitals (No Feat. Eng.)	XGBoost	0.03	0.0452	0.6299	0.0122	0.498	0.0038	0.0012	0.9969	40	12738	844	684631	0.5	0.0031	0.0059
Raw Vitals (No Feat. Eng.)	Random Forest	0.0301	0.0285	0.6279	0.0351	0.4343	0.1153	0.306	0.5175	6165	6613	209782	475693	0.5	0.4825	0.0539
Raw Vitals (No Feat. Eng.)	Logistic Regression	0.0262	0.0242	0.5942	0.0299	0.3872	0.1032	0.423	0.4373	7190	5588	289934	395541	0.5	0.5627	0.0464
Dummy Classifier	Stratified Dummy	0.0183	0.0178	0.4998	0.0178	0.4998	0.0176	0.0181	0.9824	225	12553	12399	673076	0.5	0.0176	0.0177

4-HOUR RESOLUTION																
Version	Model	AUPRC	Precision (PPV)	AUROC	F0.5	F1 Mac	F2	FPR	FNP	TP	FN	FP	TN	Threshold	Recal	F1
Stacked Ensemble	XGBoost (meta).	0.0663	0.0855	0.7289	0.0982	0.5413	0.1777	0.0666	0.7566	1193	3709	12767	179031	0.7299	0.2434	0.1265
Stacked Ensemble	XGBoost (meta)	0.0663	0.0401	0.7289	0.0495	0.3709	0.1674	0.4991	0.1850	3995	907	95723	96075	0.3934	0.815	0.0764
Stacked Ensemble	XGBoost (meta)	0.0663	0.0611	0.7289	0.0742	0.4962	0.2063	0.1992	0.4925	2488	2414	38204	153594	0.5945	0.5075	0.1091
Stacked Ensemble	Logistic Regression (meta)	0.0653	0.0816	0.7275	0.0945	0.5380	0.1796	0.0738	0.7434	1258	3644	14164	177634	0.717	0.2566	0.1238
Stacked Ensemble	Logistic Regression (meta)	0.0653	0.0399	0.7275	0.0493	0.3706	0.1667	0.4993	0.1881	3980	922	95772	96026	0.3525	0.8119	0.0761
Stacked Ensemble	Logistic Regression (meta)	0.0653	0.0608	0.7275	0.0737	0.4959	0.2049	0.1990	0.4963	2469	2433	38164	153634	0.5955	0.5037	0.1084
Engineered Features	XGBoost	0.0650	0.0541	0.7246	0.0661	0.4702	0.1985	0.2664	0.4037	2923	1979	51104	140694	0.5	0.5963	0.0992
Engineered Features	XGBoost	0.0650	0.0391	0.7246	0.0482	0.3609	0.1642	0.5194	0.1742	4048	854	99613	92185	0.3	0.8258	0.0746
Engineered Features	Random Forest	0.0602	0.0575	0.7175	0.0697	0.4925	0.1938	0.1999	0.5233	2337	2565	38335	153463	0.5	0.4767	0.1026
Engineered Features	Random Forest	0.0602	0.0431	0.7175	0.0531	0.4043	0.1751	0.4252	0.2513	3670	1232	81543	110255	0.3	0.7487	0.0815
Engineered Features	Logistic Regression	0.0553	0.0454	0.7027	0.0558	0.4274	0.1795	0.3694	0.3129	3368	1534	70856	120942	0.5	0.6871	0.0851
Engineered Features	Logistic Regression	0.0553	0.0293	0.7027	0.0363	0.2048	0.1300	0.7854	0.0732	4543	359	150638	41160	0.3	0.9268	0.0568
Raw Vitals	XGBoost	0.0399	0.0776	0.6205	0.0336	0.5020	0.0125	0.0031	0.9897	50	4822	594	188001	0.5	0.0103	0.0181
Rule-Based Baseline	TOKS Proxy	0.0387	0.0316	0.6424	0.0391	0.2853	0.1371	0.6564	0.1708	4040	832	123803	64792	3	0.829	0.061
Rule-Based Baseline	TOKS Proxy	0.0387	0.0387	0.6424	0.0476	0.3966	0.1572	0.4326	0.3266	3281	1591	81586	107009	5	0.673	0.073
Rule-Based Baseline	TOKS Proxy	0.0387	0.0428	0.6424	0.0517	0.4830	0.1388	0.1831	0.6833	1543	3329	34533	154062	7	0.317	0.075
Rule-Based Baseline	TOKS Proxy	0.0387	0.0528	0.6424	0.0545	0.5153	0.0605	0.0291	0.9372	306	4566	5494	183101	10	0.063	0.057
Raw Vitals	Random Forest	0.0387	0.0392	0.6148	0.0478	0.4597	0.1406	0.2522	0.6018	1940	2932	47571	141024	0.5	0.3982	0.0713
Raw Vitals	Logistic Regression	0.0311	0.0304	0.5665	0.0375	0.3821	0.1242	0.4474	0.4563	2649	2223	84379	104216	0.5	0.5437	0.0576
Dummy Classifier	Stratified Dummy	0.0252	0.0253	0.5001	0.0252	0.5001	0.0247	0.0245	0.9754	120	4752	4614	183981	0.5	0.0246	0.025

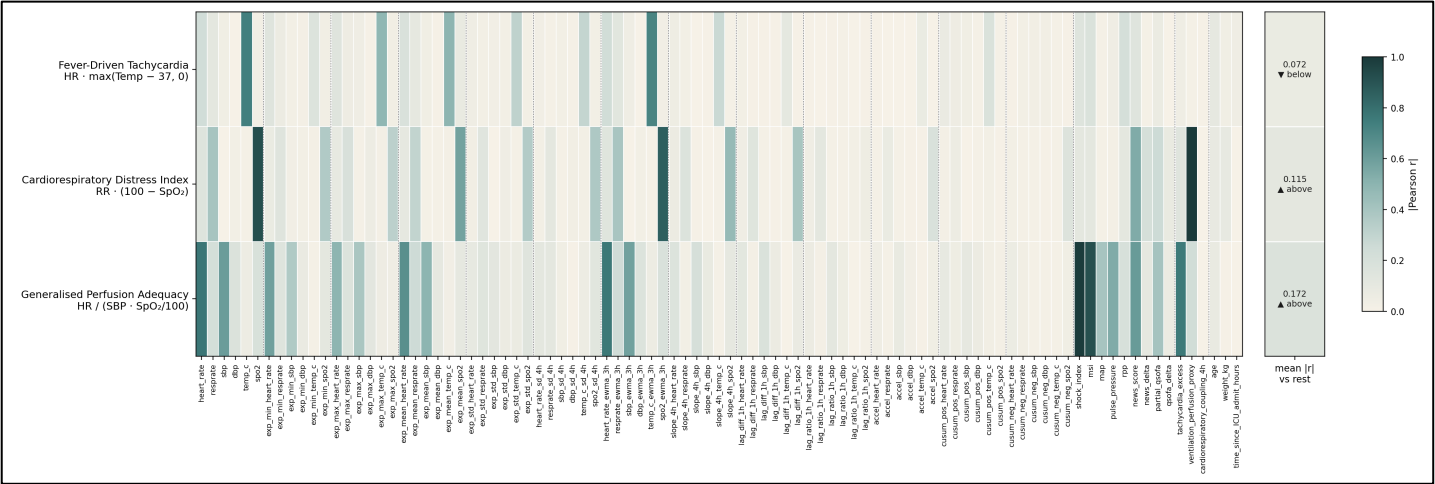
For is_sepsis_12h

15-MINUTE RESOLUTION																
Version	Model	AUPRC	Precision (PPV)	AUROC	F0.5	F1 Mac	F2	FPR	FNR	TP	FN	FP	TN	Threshold	Recal	F1
Stacked Ensemble	Logistic Regression (meta)	0.0826	0.0949	0.7362	0.1103	0.5454	0.214	0.0859	0.6882	25396	56054	242126	2575973	0.6941	0.3118	0.1455
Stacked Ensemble	Logistic Regression (meta)	0.0826	0.0448	0.7362	0.0553	0.3755	0.1835	0.498	0.1912	65874	15576	1403363	1414736	0.3386	0.8088	0.085
Stacked Ensemble	Logistic Regression (meta)	0.0826	0.0717	0.7362	0.0867	0.5045	0.2332	0.1996	0.4663	43468	37982	562579	2255520	0.5816	0.5337	0.1265
Stacked Ensemble	XGBoost (meta)	0.0838	0.0981	0.7397	0.1134	0.5482	0.2127	0.0799	0.6994	24485	56965	225166	2592933	0.718	0.3006	0.1479
Stacked Ensemble	XGBoost (meta)	0.0838	0.0455	0.7397	0.0561	0.3758	0.1862	0.499	0.1773	67009	14441	1406143	1411956	0.3764	0.8227	0.0862
Stacked Ensemble	XGBoost (meta)	0.0838	0.0723	0.7397	0.0875	0.505	0.2353	0.2	0.4607	43925	37525	563495	2254604	0.5806	0.5393	0.1275
Engineered Features (Single Model)	Logistic Regression	0.0689	0.0342	0.7195	0.0423	0.2236	0.149	0.764	0.0674	38079	2750	1076438	332507	0.3	0.9326	0.0659
Engineered Features (Single Model)	Logistic Regression	0.0689	0.0561	0.7195	0.0686	0.4553	0.2084	0.3163	0.3514	26482	14347	445636	963309	0.5	0.6486	0.1033
Engineered Features (Single Model)	Random Forest	0.0788	0.0394	0.7382	0.0487	0.302	0.1676	0.6398	0.0955	36930	3899	901425	507520	0.3	0.9045	0.0754
Engineered Features (Single Model)	Random Forest	0.0788	0.0633	0.7382	0.0772	0.4801	0.2233	0.2594	0.3947	24713	16116	365438	1043507	0.5	0.6053	0.1147
Engineered Features (Single Model)	XGBoost	0.0866	0.0773	0.7516	0.0931	0.5151	0.2411	0.1773	0.4873	20934	19895	249820	1159125	0.3	0.5127	0.1344
Engineered Features (Single Model)	XGBoost	0.0866	0.1241	0.7516	0.1294	0.555	0.1485	0.0319	0.8439	6375	34454	44986	1363959	0.5	0.1561	0.1383
Raw Vitals (No Feat. Eng.)	Logistic Regression	0.0453	0.0411	0.586	0.0505	0.4028	0.1581	0.4178	0.4539	6954	5780	162103	225914	0.5	0.5461	0.0765
Raw Vitals (No Feat. Eng.)	Random Forest	0.0516	0.0493	0.6269	0.06	0.4527	0.1737	0.2987	0.5284	6005	6729	115901	272116	0.5	0.4716	0.0892
Raw Vitals (No Feat. Eng.)	XGBoost	0.0516	0.0855	0.6283	0.0644	0.5128	0.037	0.0114	0.9676	413	12321	4415	383602	0.5	0.0324	0.047
Dummy Classifier	Stratified Dummy	0.0282	0.0284	0.5001	0.0283	0.5001	0.0281	0.0278	0.972	1144	39685	39124	1369821	0.5	0.028	0.0282

1-HOUR RESOLUTION																
Version	Model	AUPRC	Precision (PPV)	AUROC	F0.5	F1 Mac	F2	FPR	FNP	TP	FN	FP	TN	Threshold	Recal	F1
Stacked Ensemble	Logistic Regression (meta)	0.0859	0.1027	0.7374	0.118	0.5502	0.2128	0.0782	0.7093	6431	15689	56163	661651	0.707	0.2907	0.1518
Stacked Ensemble	Logistic Regression (meta)	0.0859	0.048	0.7374	0.0591	0.3779	0.1941	0.4988	0.1844	18040	4080	358078	359736	0.3396	0.8156	0.0906
Stacked Ensemble	Logistic Regression (meta)	0.0859	0.0761	0.7374	0.0919	0.5077	0.2423	0.1994	0.4668	11795	10325	143149	574665	0.5875	0.5332	0.1332
Stacked Ensemble	XGBoost (meta)	0.089	0.1077	0.741	0.1223	0.5535	0.2057	0.0679	0.7338	5889	16231	48773	669041	0.7299	0.2662	0.1534
Stacked Ensemble	XGBoost (meta)	0.089	0.0487	0.741	0.06	0.3784	0.1972	0.4994	0.1704	18351	3769	358501	359313	0.3864	0.8296	0.092
Stacked Ensemble	XGBoost (meta)	0.089	0.0763	0.741	0.0921	0.5079	0.2427	0.1992	0.4661	11810	10310	143001	574813	0.5855	0.5339	0.1335
Engineered Features (Single Model)	Logistic Regression	0.0736	0.0371	0.7228	0.0459	0.2411	0.1599	0.7408	0.0736	20491	1629	531761	186053	0.3	0.9264	0.0714
Engineered Features (Single Model)	Logistic Regression	0.0736	0.0599	0.7228	0.0732	0.4575	0.2197	0.319	0.3403	14593	7527	229002	488812	0.5	0.6597	0.1098
Engineered Features (Single Model)	Random Forest	0.0816	0.0418	0.7382	0.0517	0.3033	0.1766	0.6416	0.0911	20105	2015	460529	257285	0.3	0.9089	0.08
Engineered Features (Single Model)	Random Forest	0.0816	0.0658	0.7382	0.08	0.4796	0.2295	0.2663	0.3917	13456	8664	191163	526651	0.5	0.6083	0.1187
Engineered Features (Single Model)	XGBoost	0.0889	0.0774	0.751	0.0934	0.5092	0.2461	0.1984	0.4597	11951	10169	142422	575392	0.3	0.5403	0.1354
Engineered Features (Single Model)	XGBoost	0.0889	0.1258	0.751	0.1322	0.5562	0.1562	0.0356	0.8337	3679	18441	25574	692240	0.5	0.1663	0.1432
Raw Vitals (No Feat. Eng.)	Logistic Regression	0.044	0.0402	0.5902	0.0493	0.4027	0.1554	0.4164	0.4503	11796	9664	281810	394983	0.5	0.5497	0.0749
Raw Vitals (No Feat. Eng.)	Random Forest	0.0504	0.0471	0.6283	0.0576	0.4449	0.1707	0.3173	0.5049	10625	10835	214779	462014	0.5	0.4951	0.0861
Raw Vitals (No Feat. Eng.)	XGBoost	0.0506	0.0854	0.6283	0.0607	0.5112	0.0325	0.0096	0.9719	604	20856	6468	670325	0.5	0.0281	0.0423
Dummy Classifier	Stratified Dummy	0.0299	0.0283	0.4992	0.0283	0.4992	0.0282	0.0298	0.9718	623	21497	21379	696435	0.5	0.0282	0.0282

4-HOUR RESOLUTION																
Version	Model	AUPRC	Precision (PPV)	AUROC	F0.5	F1 Macr	F2	FPR	FNP	TP	FN	FP	TN	Threshold	Recal	F1
Stacked Ensemble	Logistic Regression (meta)	0.0865	0.0956	0.7242	0.1119	0.5361	0.2297	0.1245	0.6463	2496	4560	23616	166028	0.6642	0.3537	0.1505
Stacked Ensemble	Logistic Regression (meta)	0.0865	0.057	0.7242	0.0701	0.3852	0.2227	0.4998	0.1874	5734	1322	94775	94869	0.3466	0.8126	0.1066
Stacked Ensemble	Logistic Regression (meta)	0.0865	0.0846	0.7242	0.1015	0.5124	0.2513	0.1992	0.5051	3492	3564	37770	151874	0.6074	0.4949	0.1445
Stacked Ensemble	XGBoost (meta)	0.0868	0.1018	0.725	0.117	0.5438	0.2123	0.0956	0.7086	2056	5000	18139	171505	0.6971	0.2914	0.1509
Stacked Ensemble	XGBoost (meta)	0.0868	0.0575	0.725	0.0706	0.3857	0.2244	0.4996	0.181	5779	1277	94749	94895	0.3864	0.819	0.1074
Stacked Ensemble	XGBoost (meta)	0.0868	0.0842	0.725	0.101	0.512	0.2501	0.1993	0.5074	3476	3580	37793	151851	0.6084	0.4926	0.1439
Engineered Features (Single Model)	Logistic Regression	0.0779	0.0423	0.7038	0.0522	0.2184	0.1788	0.7829	0.0714	6552	504	148467	41177	0.3	0.9286	0.0809
Engineered Features (Single Model)	Logistic Regression	0.0779	0.0654	0.7038	0.0799	0.4451	0.2369	0.3657	0.3122	4853	2203	69348	120296	0.5	0.6878	0.1194
Engineered Features (Single Model)	Random Forest	0.0837	0.0501	0.7204	0.0618	0.3198	0.2044	0.6242	0.1145	6248	808	118376	71268	0.3	0.8855	0.0949
Engineered Features (Single Model)	Random Forest	0.0837	0.0748	0.7204	0.0906	0.4852	0.247	0.2677	0.4184	4104	2952	50764	138880	0.5	0.5816	0.1325
Engineered Features (Single Model)	XGBoost	0.0876	0.0762	0.7286	0.0924	0.4846	0.254	0.2748	0.3907	4299	2757	52109	137535	0.3	0.6093	0.1355
Engineered Features (Single Model)	XGBoost	0.0876	0.1155	0.7286	0.1221	0.5476	0.1477	0.0452	0.8413	1120	5936	8581	181063	0.5	0.1587	0.1337
Raw Vitals (No Feat. Eng.)	Logistic Regression	0.0478	0.045	0.5771	0.0551	0.4105	0.1662	0.404	0.4925	3552	3447	75337	111131	0.5	0.5075	0.0827
Raw Vitals (No Feat. Eng.)	Random Forest	0.0564	0.0564	0.6238	0.0684	0.4623	0.1882	0.2844	0.5472	3169	3830	53034	133434	0.5	0.4528	0.1003
Raw Vitals (No Feat. Eng.)	XGBoost	0.0578	0.0834	0.6313	0.0576	0.508	0.0298	0.0106	0.9743	180	6819	1978	184490	0.5	0.0257	0.0393
Dummy Classifier	Stratified Dummy	0.0359	0.0365	0.5003	0.0364	0.5003	0.0361	0.0353	0.9646	254	6802	6699	182945	0.5	0.036	0.0363
Rule-Based Baseline	TOKS Proxy	0.05194	0.04317796	0.61789	0.05324	0.295388	0.17711	0.656	0.21132	5520	1479	122243	64145	3	0.78868	0.0818736
Rule-Based Baseline	TOKS Proxy	0.05194	0.052541035	0.61789	0.06435	0.407981	0.19754	0.43122	0.36291	4459	2540	80000	160610	5	0.63709	0.097076176
Rule-Based Baseline	TOKS Proxy	0.05194	0.058099568	0.61789	0.06926	0.49212	0.16357	0.18223	0.70053	2096	4903	33980	152488	7	0.29947	0.09731863
Rule-Based Baseline	TOKS Proxy	0.05194	0.065748331	0.61789	0.0716	0.517846	0.09767	0.05929	0.88884	778	6221	11055	1754319	9	0.11116	0.082625319

Appendix 4 - Novel Features Pairwise Correlation



Each row shows $|r|$ between the feature and every other feature. Mean of mean $|r| = 0.107$, median = 0.115